# EduTutor & EduSchool

## System Design Document

Version 10.0 — June 2026

| Product | For | Accent |
| --- | --- | --- |
| EduTutor | Independent tutors & their learners / parents | Pink |
| EduSchool | Schools, teachers, learners & parents | Green |
| Shared Core | The engine powering both products | Blue |

# 1. The Problem We Are Solving

> ***"Sending a child to school does not mean they will learn. Giving a learner access to content does not mean they will use it, or use it correctly."***

Education has a delivery problem and an accountability problem. Both have existed for decades and neither has been fully solved by any platform currently available in South Africa or globally.

## 1.1 The Delivery Problem

Teachers and tutors spend the majority of their time doing something a recording can do: explaining the same concept, to different learners, over and over again. Every time a new class starts, the explanation starts again. Every tutor session begins with re-teaching what was not retained. This is an enormous waste of skilled human time.

Meanwhile, learners wait. They wait for the teacher to get to their question. They wait for their marked test to come back. They wait for a tutor session. In that waiting, momentum is lost and gaps grow wider.

## 1.2 The Accountability Problem

Every existing platform — Siyavula, Khan Academy, HelloAida, FoondaMate — solves the content delivery problem to varying degrees. What none of them solve is the accountability and intervention loop.

| Platform | What It Does Well | What It Does Not Solve |
|---|---|---|
| Siyavula | Free adaptive Maths & Science practice. CAPS-aligned. Zero-rated. | Maths & Science only. No tutor or teacher monitoring. No alert when learner is struggling. No parent visibility. Learner must choose to open it. |
| Khan Academy | Broad free content library. Self-paced video lessons. | No managed learning environment. No one is notified when a learner disengages. Completion ≠ understanding. |
| HelloAida | AI tutoring chatbot. SA-built. Multilingual. CAPS & IEB. | Reactive only — learner must ask a question. No teacher dashboard. No class management. No parent portal. No video lessons. |
| FoondaMate / Sammo | WhatsApp-based. Data-light. Accessible to low-income learners. | Chatbot answers only. No monitoring. No structured content path. No intervention workflow. |
| SubSchool | Video-to-course AI. Closest global equivalent to our model. | Not SA-specific. No class management. No parent portal. No struggling learner alerts. |

The gap is not content. The gap is the 'accountability layer' — the closed loop between learning happening, the right person being notified, that person being able to act, and the action being tracked. No platform in the South African market has built this loop completely.

# 2. Our Solution

EduTutor and EduSchool are not content platforms. They are "managed individualised learning systems." The content, the AI, the human educator, and the data all work together in a closed loop that ensures learning is actually taking place — and that when it is not, someone is notified and can act.

**We are not just giving learners content. We are changing how learning is monitored, how educators teach, and how quickly problems are identified and solved.**

### 2.1 The Core Accountability Loop

Every interaction on both platforms feeds a live accountability loop with four stages:

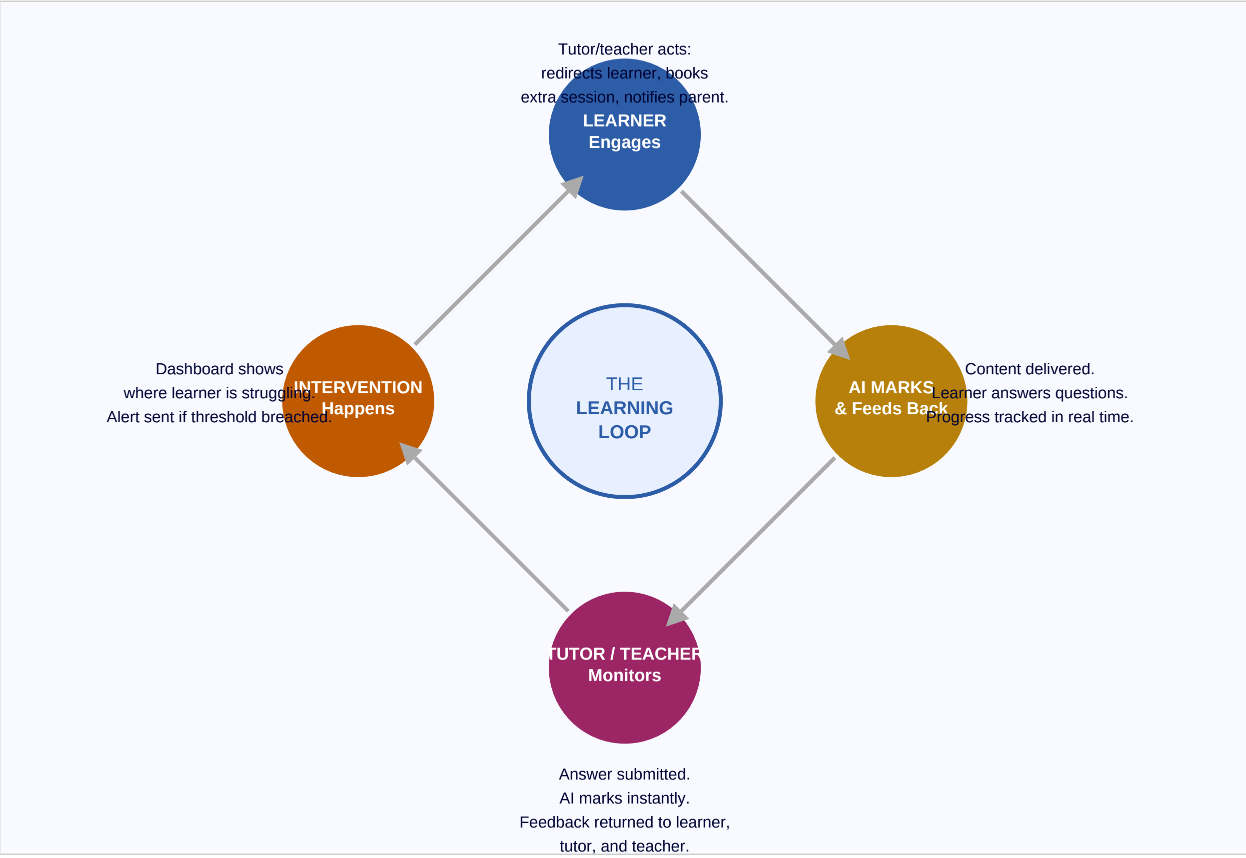



*Figure 1: The Core Accountability Loop — the engine behind both products*

| Stage | What Happens | Who Benefits |
|---|---|---|
| 1. Learner Engages | Learner watches a lesson and answers practice questions. Progress is tracked in real time as it happens. | Learner gets structured, self-paced content with immediate feedback. |
| 2. AI Marks & Feeds Back | Answer is submitted. AI marks it instantly, identifies the exact error, explains the correction, and recommends a lesson to revisit. | Learner gets feedback in seconds, not days. Tutor/teacher gets data without marking. |
| 3. Tutor / Teacher Monitors | Dashboard is updated live. If a learner falls below a threshold or fails consolidation repeatedly, an alert is triggered. | Educator sees exactly where each learner is struggling before the next session begins. |

| | | |
|---|---|---|
| 4. Intervention Happens | Educator redirects the learner to a specific lesson, sends a message, books an extra session, or notifies the parent. | Learner gets targeted help at the right time. Problems are caught in weeks, not at the end of term. |

## 2.2 What This Changes for Each Person

| Person | Before | After |
|---|---|---|
| Teacher | Spends most of class time re-explaining concepts. Marks 35 scripts over the weekend. Finds out a learner is failing at the end-of-term exam. | Records a lesson once. AI handles marking. Dashboard flags a struggling learner in week two. Teacher focuses on planning, recording, and targeted support. |
| Tutor | Starts every session by re-teaching. Manually tracks what each learner covered. Has no data between sessions. | Opens the dashboard before the session. Sees exactly what the learner struggled with this week. Session is focused before it starts. |
| Learner | Waits for feedback. Gets stuck and stops. Doesn't know what they don't know. | Gets feedback in seconds. Step-by-step hints help self-correction. Cannot progress without demonstrated understanding. |
| Parent | Waits for a school report or relies on the child's own account of how things are going. | Sees last night's activity, the AI feedback report, and the topic their child is stuck on — in real time. |

## 2.3 Two Products, One Engine

The platform is built as two separate deployable products — EduTutor and EduSchool — with different backends, different billing models, and different interfaces. Underneath both sits the same shared engine: the same AI marking system, the same content delivery infrastructure, and the same learner progress tracking.

| | EduTutor | EduSchool |
|---|---|---|
| Sold To | Independent tutors & parents of learners | Schools as an institution |
| Sign-up | Self-service online. Card payment. Active same day. | Institutional contract. School Admin created by platform. |
| Billing | Tutor monthly subscription + learner tier | Per-school based on learner count |
| Content Ownership | Tutor-owned. Private by default. | Teacher-owned. School-wide or platform-wide sharing. |
| Monitoring | Tutor monitors individual learners | Teacher monitors entire class. School Admin sees school-wide. |
| Parent Access | Optional (future feature) | Included. Teacher invites parent. Read-only portal. |
| Backend | Separate database & billing | Separate database & billing |

| | | |
|---|---|---|
| Shared Engine | AI marking, content delivery, learner flow | Same |

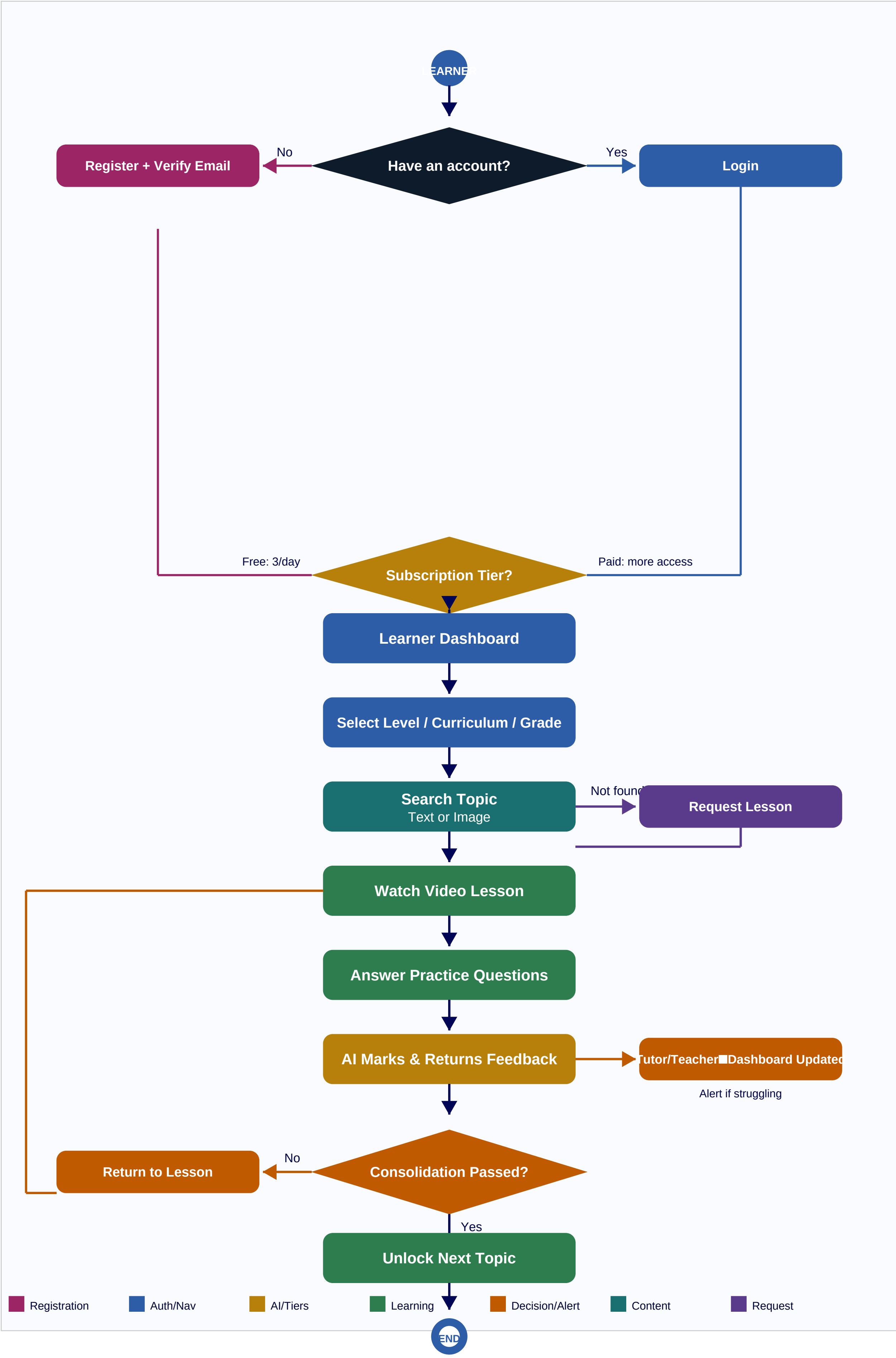
## 3. Shared Core — What Both Products Are Built On

The following components are identical across both products. They are built once and shared. When the AI marking improves, both products benefit immediately.

| Component | Description |
|---|---|
| Learner Registration & Auth | Self-registration with email verification. Login, password reset, profile settings including level/curriculum/grade changes. |
| Content Engine | Dynamic content hierarchy: Curriculum → Grade → Subject → Topic → Lesson → Question. Nothing hardcoded. Everything created by admin, tutors, or teachers. |
| Video Lesson Delivery | Lessons uploaded and streamed. Linked to topics. Sequenced into learning paths. Learner tracks completion. |
| AI Marking Engine | Answer submitted → OCR if handwritten → packaged with question, model answer, and rubric → sent to AI API → mark + error location + correction + next step returned instantly. |
| Consolidation Gating | Every topic ends with a consolidation exercise. Learner cannot progress without passing. Failed questions direct back to the relevant lesson. |
| Step-by-Step Hints | Steps are hidden by default. The learner attempts the question independently first. If stuck, the learner can choose to reveal Step 1, then further steps one at a time. Full solution only after submission. |
| Learner Dashboard | Progress per subject, recent activity, performance summary, lesson requests, subscription status. |
| Learner Subscription Tiers | Free (3 lessons/3 answers per day), Basic (unlimited lessons/20 answers), Standard (40), Premium (80). All configurable by admin. |
| Platform Admin | Full control: users, content, AI settings, tier pricing, feature flags, branding. All without code changes. |

## 3.1 Learner Flow Diagram

The learner journey is identical on both EduTutor and EduSchool. The monitoring alert to the right of the AI marking step shows where the tutor/teacher dashboard is updated in real time.



EARNE
Have an account?
No
Register + Verify Email
Yes
Login
Free: 3/day
Subscription Tier?
Paid: more access
Learner Dashboard
Select Level / Curriculum / Grade
Search Topic
Text or Image
Not found
Request Lesson
Watch Video Lesson
Answer Practice Questions
AI Marks & Returns Feedback
Tutor/Teacher■Dashboard Updated
Alert if struggling
Consolidation Passed?
No
Return to Lesson
Yes
Unlock Next Topic
END
Registration
Auth/Nav
AI/Tiers
Learning
Decision/Alert
Content
Request

*Figure 2: Learner flow — applies to both EduTutor and EduSchool*

PRODUCT 1

# EduTutor

For independent tutors and their learners. Self-signup. Tutor workspace. Real-time learner monitoring.

# T1. EduTutor Overview

EduTutor is sold directly to independent tutors. A tutor signs up, chooses a subscription package, and immediately has a private workspace to create content and manage their learners. Parents subscribe separately for their child's access. The tutor's dashboard gives them live visibility into every learner's progress so that every session is data-driven rather than reactive.

**EduTutor turns every tutor into a data-informed educator — not just someone who explains things, but someone who knows exactly what each learner needs before the session begins.**

# T2. Tutor Registration & Packages

## T2.1 Registration Flow

1. Tutor visits EduTutor and selects "Register as a Tutor."
2. Enters name, email, password, subject specialisation(s), and a brief bio.
3. Email verification link sent. Account inactive until verified.
4. Tutor selects a subscription package and completes payment.
5. Dashboard is activated immediately on payment confirmation.

## T2.2 Tutor Subscription Packages — Seat-Based Pricing

Tutors pay based on how many active learners they have in their space. There are no hard content upload caps — the tutor pays for seats, and each seat comes with a defined daily experience for the learner that mirrors the equivalent learner tier. The learner pays nothing to the platform — their access is covered by the tutor.

| Package | Active Learner Seats | Monthly Cost | Learner Gets (per day) | Equivalent Learner Tier |
|---|---|---|---|---|
| Solo | 1 – 5 learners | R199/month | Unlimited lessons. 20 AI answers/day. | Basic |
| Small | 6 – 15 learners | R349/month | Unlimited lessons. 40 AI answers/day. | Standard |
| Medium | 16 – 30 learners | R599/month | Unlimited lessons. 80 AI answers/day. | Premium |
| Large | 31 – 50 learners | R899/month | Unlimited lessons. 80 AI answers/day. | Premium |
| Studio | 51 – 100 learners | R1,299/month | Unlimited lessons. 80 AI answers/day. | Premium |
| Enterprise | 100+ learners | Custom pricing | Custom limits. | Custom |

**The tutor buys seats. Each seat covers a learner's full access within that space. Learners pay nothing to the platform. Better packages give learners higher daily limits.**

### T2.3 How Learner Access Works Inside a Tutor Space

A learner invited by a tutor has two contexts on the platform. Their limits depend on which context they are operating in:

| Context | Limits Applied | Who Pays |
|---|---|---|
| Inside the tutor's space | Limits set by the tutor's package (e.g. 20/40/80 AI answers/day). Unlimited access to the tutor's content, lessons, flashcards, and sessions. | Tutor's subscription covers the learner. Learner pays nothing. |
| Outside the tutor's space (independent study) | Reverts to the learner's own account tier. If learner has no paid tier of their own: Free tier (3 lessons / 3 answers/day). If learner has their own paid plan: that plan's limits apply. | Learner's own subscription (or Free tier). |

A learner on Free who has a tutor gets the best of both: Free limits for independent study, and their tutor's package limits when working inside the tutor's space. The platform automatically applies the correct limit based on context.

### T2.4 Seat Usage Rules

- A seat is counted as active when a learner has accepted an invitation and logged in at least once in the current billing period.
- A learner who has not been active in 60 days is marked inactive and their seat is released automatically. The tutor is notified before release.
- When a tutor reaches 80% of their seat limit, they receive an upgrade prompt.
- When a tutor reaches 100% of their seat limit, new invitations are blocked until they upgrade or a seat is freed.
- Upgrading takes effect immediately. Downgrading takes effect at the end of the billing period.
- A tutor cannot downgrade if doing so would put them over the new seat limit. They must remove learners first.

### T2.5 Tutor Billing Dashboard

| Dashboard Section | Content |
|---|---|
| Current Package | Package name, monthly cost, renewal date. |
| Seat Usage | Active learners vs seat limit (e.g. 12/15 seats used). Visual progress bar. |
| Cost Per Learner | Monthly cost divided by active learners (e.g. R349 / 12 = R29.08/learner). |
| Learner Experience | Current AI answer limit per learner per day and lesson access level. |
| Upgrade Preview | What the next package costs, how many extra seats it unlocks, and what improved limits it gives each learner. |

| | |
|---|---|
| Upgrade Prompt | Appears at 80% seat capacity with one-tap upgrade action. |

## T3. Tutor Dashboard & Monitoring

The tutor dashboard is designed around one question: “what does each learner need from me right now?” Every data point shown feeds directly into session planning and intervention.

| Dashboard Section | Content |
|---|---|
| Monitoring Feed | Real-time activity: which learner answered what, scored how much, and passed or failed consolidation. Struggling learner flags appear here instantly. |
| My Learners | Full list of assigned learners with progress % per topic, last active time, and performance trend. |
| Session Prep | Per-learner summary of what they struggled with since the last session. Tutor walks into every session knowing exactly what to focus on. |
| AI Feedback Reports | Every submitted answer and the AI’s marking. Tutor can override any mark and add a manual comment. |
| My Content | All created curricula, topics, lessons, and questions. Upload count vs package limit. |
| Lesson Requests | Requests submitted by learners for topics not yet covered. |
| Billing & Package | Current package, usage stats, upgrade/downgrade options. |

## T4. Content Creation

1. Navigate to “My Content” and create a Curriculum (e.g. CAPS Mathematics Grade 11).
2. Add a Grade, Subject, and Topic within the curriculum.
3. Upload a Video Lesson: title, video file, description. Upload count checked against package limit.
4. Add Practice Questions. Each requires: question text, model answer, and marking rubric for AI.
5. Mark specific questions as the Consolidation Exercise for the topic.
6. Publish the topic. Visible to assigned learners immediately.

Content stays in draft until published. Draft content is visible only to the tutor and admin. A question cannot be published without a model answer and rubric — these are required for AI marking.

## T5. Learner Management & Intervention

### T5.1 Inviting Learners

1. Navigate to “My Learners” and invite by email. Learner count checked against package limit.
2. Learner receives invitation and registers or links an existing account to the tutor.
3. Tutor assigns learner to a curriculum, grade, and starting topic.

### T5.2 Intervention Workflow

When a learner is flagged as struggling (configurable threshold — e.g. fails consolidation twice, or scores below 50% on three consecutive questions), the tutor is alerted. The tutor can then:

- Redirect the learner to a specific lesson from within the platform.
- Send a message or note to the learner explaining what to revisit.
- Plan the next session around the flagged topic — session prep view shows this automatically.
- Override the AI mark if the feedback was incorrect and add a manual correction.

Every intervention action is logged. Tutor can see the full history of alerts and responses per learner.

# T6. EduTutor Flow Diagram

Complete tutor workflow: registration, package selection, content creation, real-time learner monitoring, and the intervention loop.

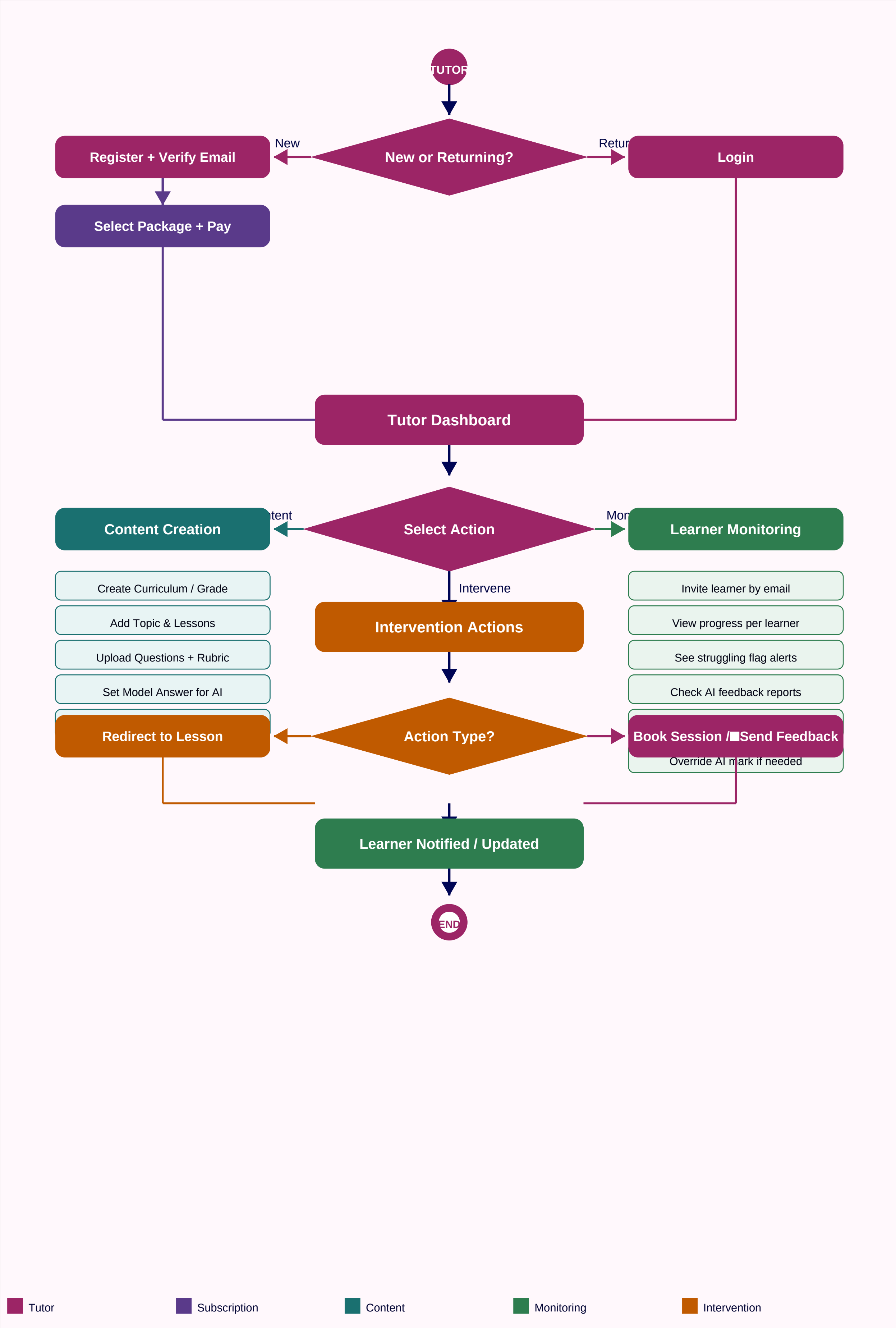

TUTOR
New or Returning?
New
Register + Verify Email
Select Package + Pay
Retur
Login
Tutor Dashboard
Select Action
tent
Content Creation
Create Curriculum / Grade
Add Topic & Lessons
Upload Questions + Rubric
Set Model Answer for AI
Mon
Learner Monitoring
Invite learner by email
View progress per learner
See struggling flag alerts
Check AI feedback reports
Override AI mark if needed
Intervene
Intervention Actions
Action Type?
Redirect to Lesson
Book Session /■Send Feedback
Learner Notified / Updated
END
Tutor
Subscription
Content
Monitoring
Intervention

*Figure 3: EduTutor workflow*

PRODUCT 2

# EduSchool

For schools, teachers, learners, and parents. Institutional. Class-based. Lesson sharing. Parent portal.

## S1. EduSchool Overview

EduSchool is sold to schools as an institutional product. It extends the core platform with class-based management, a lesson sharing library between teachers, a parent portal, and a school admin layer. The central insight is that a teacher’s most valuable time should not be spent delivering the same explanation twice.

**A lesson recorded once by one teacher can be delivered to every class, every year, forever. The teacher’s time is freed for what only a human can do: identify who is struggling, intervene early, and support the learner directly.**

## S2. Account Structure

| Account | Created By | Scope & Purpose |
|---|---|---|
| School Admin | Platform Admin (on school sign-up) | Manages the entire school. Creates teacher accounts. Views all classes and learners. Approves platform-wide lesson sharing. Manages school subscription and billing. |
| Teacher | School Admin | Manages one or more classes. Creates, records, and assigns lessons. Views class monitoring dashboard. Shares lessons with colleagues. Manages parent access for their class. |
| Learner (School) | Teacher or School Admin | Same learner experience as the standard platform. Enrolled in a class. Progress visible to their teacher, school admin, and linked parent in real time. |
| Parent | Teacher (sends invite to parent’s email) | Read-only portal. Sees their child’s progress, scores, AI feedback reports, and teacher notes. Cannot see other learners or modify anything. |

## S3. School Subscription Model — Flat Per-Learner Pricing

EduSchool pricing is a flat rate per enrolled learner per month: R28/learner/month, with a 60 AI-answer-per-day limit per learner (reduced from 80 to maintain margin — still 1.5x the Standard self-managed tier). This single rate guarantees at least 55% gross margin at any scale, from a 50-learner school to a 2,000-learner district. Feature tiers (advanced reporting, dedicated account management, custom branding) are unlocked at enrolment thresholds at no extra per-learner cost — they are operational commitments, not infrastructure costs, so they do not affect the margin calculation.

| Enrolment Size | Rate | Monthly Fee Example | Margin | Included Feature Tier |
|---|---|---|---|---|
| Up to 100 learners | R28/learner | 100 × R28 = R2,800 | ~57% | Core: full platform, parent portal, class dashboards, AI marking, lesson sharing. |

| | | | | |
|---|---|---|---|---|
| 101 – 300 learners | R28/learner | 200 × R28 = R5,600 | ~57% | All Core. Advanced reporting. Priority support. |
| 301 – 750 learners | R28/learner | 500 × R28 = R14,000 | ~57% | All above. Custom branding. Dedicated account manager. |
| 751 – 1,500 learners | R28/learner | 1,000 × R28 = R28,000 | ~57% | All above. SLA agreement. Onboarding support. |
| 1,501 – 2,000 learners | R28/learner | 1,800 × R28 = R50,400 | ~57% | All above. Dedicated infrastructure. |
| 2,000+ learners | Negotiated, min R28 | Custom contract | ≥55% | White-label. SA EMIS / D6 integration. Custom contract. |

**R28 per learner per month, every learner, every school, no exceptions. The bill is always: enrolled learners × R28. Feature tiers unlock automatically as enrolment grows — they cost the school nothing extra because they are not infrastructure-driven costs.**

## S3.1 Margin Verification

Infrastructure cost per learner at 60 AI answers/day is approximately R12.00/month (interpolated between the 40-answer Standard cost of R8.46 and the 80-answer Premium cost of R15.46). Tooling cost share decreases as enrolment grows. EduSchool incurs no payment gateway fee (manual EFT invoicing — see Section 26.3).

| Enrolment | Infra Cost/Learner | Tooling Share | Total Cost | Revenue | Margin |
|---|---|---|---|---|---|
| 50 learners | R12.00 | R0.101 | R12.10 | R28.00 | 56.8% |
| 200 learners | R12.00 | R0.025 | R12.03 | R28.00 | 57.0% |
| 800 learners | R12.00 | R0.006 | R12.01 | R28.00 | 57.1% |
| 2,000 learners | R12.00 | R0.003 | R12.00 | R28.00 | 57.1% |

Margin is consistently above 55% at every scale because infrastructure cost is the dominant factor and is essentially flat per learner regardless of school size. There is no underwater band — every enrolment size is profitable from the first learner.

## S3.2 Pricing Implications for Sales

This represents a meaningful increase from earlier indicative pricing (previously R12–R18/learner depending on band). A few things to keep in mind when positioning this with schools:

- R28/learner/month = R336/learner/year. For context, many private SA schools spend significantly more per learner annually on textbooks alone.
- The 60 AI-answer/day limit is generous — it covers roughly 2 questions per subject per day across a typical timetable, with room for consolidation exercises.
- Position the per-learner rate against the value delivered: AI marking that would otherwise require additional teaching staff, parent communication that would otherwise require admin time, and lesson content that does not need to be repeated every year.

- For very large districts (2,000+ learners) where R28/learner becomes commercially difficult, a custom contract can explore a lower per-learner rate only if paired with a reduced AI answer limit (e.g. 40/day) to preserve the margin floor — never reduce price without reducing the corresponding cost driver.

### S3.3 Enrolment Changes Mid-Term

- Billing is based on actual enrolled learner count each month — no fixed bands to move between. The bill simply scales: enrolled learners × R28.
- If enrolment grows mid-month, the additional learners are billed pro-rata from their activation date.
- If enrolment drops (e.g. end of school year), the next month's bill reflects the lower count automatically.
- School Admin sees a live running total on the billing dashboard: current enrolled learners × R28 = projected next bill.

## S4. Teacher Workflow

### S4.1 Lesson Management — Record Once, Teach Forever

1. Teacher records or uploads a video lesson and links it to a subject, grade, and topic.
2. Teacher browses the shared lesson library — lessons shared by colleagues in the school or platform-wide.
3. Teacher previews a shared lesson. If suitable, assigns it to their class without recording anything.
4. Teacher sequences lessons into a topic flow and assigns to a class. Learners see it immediately.

A teacher who records a brilliant explanation of trigonometry never has to deliver that explanation again. That recording serves every class, every year. Their freed time goes toward monitoring, intervention, planning, and recording new content.

### S4.2 Lesson Sharing Permissions

| Level | Visible To | Set By |
|---|---|---|
| Private | Only the teacher who created it. | Teacher |
| School-Wide | All teachers and learners within the school. | Teacher or School Admin |
| Platform-Wide | All EduSchool teachers and learners. | Teacher (with School Admin approval) |

### S4.3 Class Monitoring Dashboard

The class dashboard answers one question at a glance: "who needs my attention right now?"

| View | What the Teacher Sees |
|---|---|
| Class Completion | Every learner in the class. Completion % per topic. At a glance, who is behind. |
| Lesson Completion | Which learners have watched each assigned lesson. Time spent. Drop-off points. |
| Question Performance | Per-question scores across the class. Average. Distribution. Who scored what. |

| | |
|---|---|
| Common Mistakes | AI-aggregated summary of the most frequent errors across all learner submissions for each question. Teacher sees this before marking a single script. |
| Struggling Learners | Auto-flagged: learners below a configurable threshold (e.g. below 50%, or failed consolidation twice). Appears at the top of the dashboard. |
| Activity Feed | Real-time: lesson watched, question answered, consolidation passed or failed. |

### S4.4 Intervention

When a learner is flagged, the teacher acts from the dashboard without leaving the platform:

- Send a nudge notification to the learner.
- Assign a remedial lesson directly to the learner.
- Add a private note to the learner's profile.
- Notify the parent via the parent portal.
- Override an AI mark and add a manual correction comment.

## S5. Parent Portal

The parent portal closes the final gap: parents who currently rely on a report card once a term now have a real-time window into their child's daily learning. This is not a communication tool — it is a visibility tool.

### S5.1 Access

1. Teacher enters parent's email in the class roster.
2. Parent receives an invitation link and registers a parent account.
3. Parent account is linked to their child's learner profile only.

### S5.2 What Parents See

| Section | Content |
|---|---|
| Progress Overview | Subjects enrolled. Topics completed vs total. Overall completion %. |
| Recent Activity | Last 7 days: lessons watched, questions answered, scores received. |
| Performance by Topic | Score summary per topic. Green/amber/red indicators. |
| AI Feedback Reports | The same reports the learner receives. Parent sees exactly where their child made errors and what the correction was. |
| Consolidation Results | Which topics the child has passed and which are still in progress. |
| Teacher Notes | Notes the teacher has marked as parent-visible. |

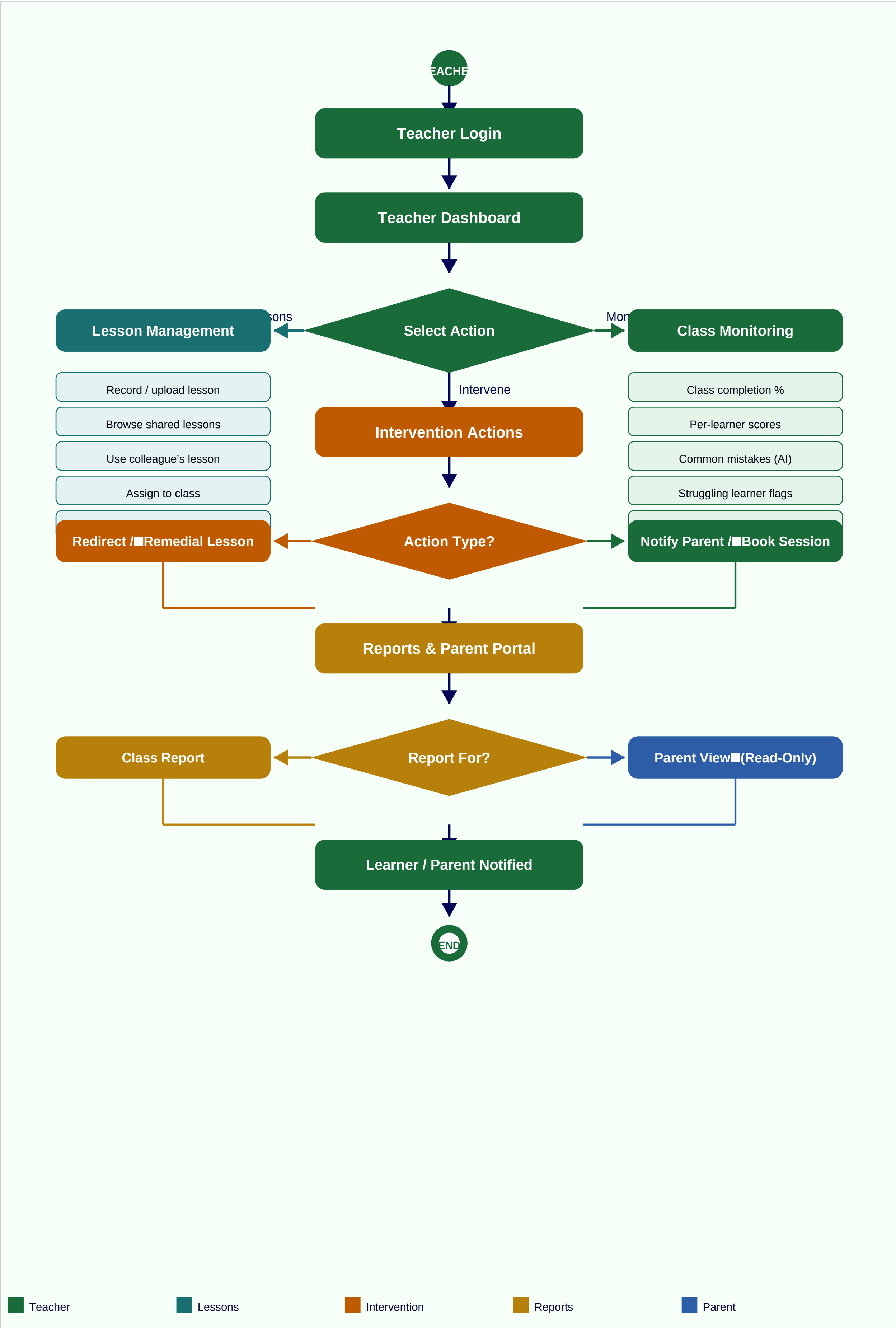
Parents cannot modify content, contact teachers through the portal, or see any other learner's data.

## S6. School Admin

| Action | Description |
| --- | --- |
| Create Teacher Accounts | Add teachers to the school. Assign to subjects and grades. |
| View All Classes | School-wide visibility across all teachers and classes. |
| View All Learner Progress | School-wide performance view by grade, subject, or class. |
| Approve Platform-Wide Sharing | Review and approve lessons teachers want to share platform-wide. |
| Manage School Subscription | Upgrade, downgrade, or cancel the school's plan. Manage billing. |
| Content Permissions | Control whether teachers can use platform-wide lessons from outside the school. |
| School Reports | Generate summary reports by grade, subject, or class for school management or the DBE. |

## S7. EduSchool Teacher Flow Diagram

Complete teacher workflow: login, lesson management including shared lessons, class monitoring dashboard, intervention actions, reporting, and parent portal.



EACHE
Teacher Login
Teacher Dashboard
Select Action
sons
Mon
Lesson Management
Class Monitoring
Record / upload lesson
Browse shared lessons
Use colleague's lesson
Assign to class
Intervene
Intervention Actions
Class completion %
Per-learner scores
Common mistakes (AI)
Struggling learner flags
Redirect /■Remedial Lesson
Action Type?
Notify Parent /■Book Session
Reports & Parent Portal
Class Report
Report For?
Parent View■(Read-Only)
Learner / Parent Notified
END
Teacher
Lessons
Intervention
Reports
Parent

*Figure 4: EduSchool teacher workflow*

# 4. AI Marking Engine

AI marking is the engine that makes the accountability loop possible at scale. Without it, a tutor managing 50 learners or a teacher with 35 students in a class cannot review every answer in real time. With it, every answer is reviewed, every error is identified, and every learner gets feedback before the next session.

## 4.1 Marking Flow

1. Learner submits an answer (typed or photo of handwritten working).
2. If a photo: OCR pre-processing extracts the text and working.
3. Answer, question text, model answer, and marking rubric are packaged into a structured prompt.
4. Prompt is sent to the AI API (provider and model configurable by admin).
5. AI returns: mark awarded, step where error occurred, corrective explanation, recommended lesson.
6. Response parsed and displayed to learner instantly. Tutor/teacher dashboard updated.

## 4.2 What the Learner Receives

- Mark awarded (e.g. 3/5).
- Clear explanation of what was correct.
- Exact step or line where the error occurred.
- Corrective explanation of the right approach.
- Recommended lesson or worked example to revisit.

## 4.3 Configuration (Admin-Managed, No Code Required)

| Config Item | Description |
|---|---|
| AI Provider | API provider (e.g. Anthropic Claude, OpenAI). Switchable from admin panel. |
| Model Version | Specific model string. Updatable by admin without deployment. |
| Prompt Template | Base marking prompt. Editable by admin to improve accuracy over time. |
| Fallback Behaviour | If API unavailable: queue for manual review or notify learner with an explanation. |
| Cost Controls | Token limits per request. Monthly spending cap. Usage dashboard for admin. |
| Override Logging | Every AI mark overridden by a tutor or teacher is logged. Admin can identify questions where AI consistently underperforms. |

# 5. Design Principles

| Principle | Description |
|---|---|
| Learning Must Be Verified | Progress means demonstrated understanding, not just content consumption. Consolidation gating ensures this. |

| | |
|---|---|
| Real-Time Accountability | Educators see what is happening as it happens. Alerts fire when a learner is struggling. Problems are caught in days, not at end of term. |
| Educator Time is Precious | AI handles marking. Lessons are recorded once. Educator time is preserved for planning, intervention, and human connection. |
| Individualised at Scale | Every learner gets a personalised feedback report on every answer. Monitoring is per-learner even in a class of 35. |
| Record Once, Deliver Forever | A lesson recorded by one educator can be used by any class, any year, on any compatible curriculum. |
| Dynamic Content | All curricula, grades, topics, and questions are created and managed without code changes. |
| Freemium Entry Point | Free tier lowers the barrier to entry for learners. Paid tiers unlock full access. |
| Separate Products, Shared Engine | EduTutor and EduSchool are separate deployments with separate data. The AI and content engine is shared. |
| Admin Configurability | Pricing, AI settings, feature flags, and branding are all admin-managed. No developer needed for configuration. |

## 6. Competitive Positioning

The South African EdTech market has strong content platforms and strong chatbot tools. Neither category has built the full accountability loop. That is the white space both products occupy.

| Capability | EduTutor | EduSchool | Siyavula | HelloAida | Khan Academy |
|---|---|---|---|---|---|
| Self-paced lessons + questions | ✓ | ✓ | ✓ (Maths/ Science) | ✓ (chat only) | ✓ |
| AI marking with error feedback | ✓ | ✓ | Adaptive hints | ✓ (chat) | ✘ |
| Tutor / teacher monitoring dashboard | ✓ | ✓ | ✘ | ✘ | ✘ |
| Struggling learner alerts | ✓ | ✓ | ✘ | ✘ | ✘ |
| Intervention workflow | ✓ | ✓ | ✘ | ✘ | ✘ |
| Parent real-time portal | ✘ (future) | ✓ | ✘ | ✘ | ✘ |
| Lesson sharing between educators | ✓ (optional) | ✓ | ✘ | ✘ | ✘ |
| Any subject / curriculum | ✓ | ✓ | ✘ (Maths/Science) | ✓ | ✓ |
| SA-built and SA-priced | ✓ | ✓ | ✓ | ✓ | ✘ |

## 8. Infrastructure, AI & Cost Architecture

This section documents every third-party service the platform depends on, the rationale for choosing it, its limitations, and the per-learner monthly cost at each subscription tier. All figures are in USD and ZAR (at R18.40/USD, June 2026).

### 8.1 AI Marking — DeepSeek V4 Flash

DeepSeek V4 Flash is the recommended AI provider for marking at launch. It is the cheapest frontier-class API available and produces strong results for structured marking tasks like question-and-answer evaluation.

| Item | Detail |
|---|---|
| Provider | DeepSeek (platform.deepseek.com) |
| Model | DeepSeek V4 Flash (non-thinking mode for speed; thinking mode available for complex questions) |
| Input cost | $0.14 per million tokens (cache miss) / $0.0028 per million tokens (cache hit — 98% cheaper when prompt prefix is reused) |
| Output cost | $0.28 per million tokens |
| Context window | 1 million tokens |
| Free grant | 5 million tokens on signup, valid 30 days. No credit card required. Enough for ~2,500–5,000 test marking calls. |
| Integration | Standard REST API. Same interface as OpenAI. Drop-in replacement. Provider is configurable by admin — switch to Claude or Gemini without code changes. |

### 8.2 DeepSeek Limitations You Must Plan For

| Limitation | Impact | Mitigation |
|---|---|---|
| Data privacy — Chinese company | Learner answers sent to DeepSeek servers in China. POPIA requires knowing where personal data is processed. Significant risk for EduSchool (minors' data). | For MVP: include in privacy policy and terms. Medium-term: self-host an open-source model (DeepSeek-R1 weights are public) on a South African or EU server. Or switch to Anthropic Claude which has EU data residency options. |
| No image/vision support on V4 Flash | Learner uploads a photo of handwritten work — DeepSeek Flash cannot read it directly. | Run OCR first. Google Vision API: free for first 1,000 images/month, then $1.50 per 1,000. Tesseract (open source, free, self-hosted) works for printed text but struggles with messy handwriting. |
| No fine-tuning | Cannot train the model on your specific rubrics or SA curriculum style. | Compensate with detailed prompt templates. Admin can edit the system prompt from the admin panel without code changes. |

| | | |
|---|---|---|
| API availability / rate limits | DeepSeek occasionally goes down or throttles. If the API is unavailable, learners get no feedback. | Implement a queue: if API fails, answer is queued and feedback delivered within 5 minutes. Fallback provider (e.g. Gemini Flash) can be configured by admin. |
| Output language | Defaults to English. SA learners may need Afrikaans or isiZulu feedback. | Instruct the model in the system prompt to respond in the learner's selected language. Works reasonably well for Afrikaans. Less reliable for isiZulu — test carefully. |

### 8.3 Video Storage & Streaming — Bunny.net Stream

Bunny.net Stream is the recommended video hosting and delivery service. It provides a full API for uploading and streaming, automatic transcoding to multiple quality levels, an embeddable player, and pay-as-you-go pricing with no minimums. It is significantly cheaper than Mux, Wistia, Vimeo, or Cloudflare Stream.

| Item | Detail |
|---|---|
| Provider | Bunny.net Stream (bunny.net) |
| Storage cost | $0.005 per GB/month (approximately R0.09/GB/month) |
| CDN delivery cost | $0.01 per GB delivered (Africa/SA region). Transcoding included free. |
| Player | Embeddable HLS player included free. Supports adaptive bitrate (auto quality for learner's connection). |
| API | Full REST API. Upload via API, receive a video ID, embed or stream via URL. No manual link management needed. |
| Upload flow | Tutor/teacher uploads video in your app → your backend calls Bunny API → Bunny stores and transcodes → video ID saved to your database → learner streams via embedded player. |
| Free tier | No permanent free tier. Pay-as-you-go from first byte. Minimum spend is negligible at small scale. |
| Note on Screencastify | Screencastify is a screen recording tool, not a video host. Use it to record lessons, then upload the resulting file to Bunny via your platform. |

Video quality tiers recommendation: serve 480p on Free learner tier, 720p on Basic/Standard, 1080p on Premium. This controls delivery bandwidth costs and prevents Free tier from consuming disproportionate infrastructure.

### 8.4 Image Storage & OCR

| Use Case | Service | Cost |
|---|---|---|
| Learner answer photos (handwritten work uploads) | Cloudinary (free tier to start) | Free tier: 25 credits/month covering ~25GB storage and 25GB bandwidth. Sufficient for MVP. Paid plan: $89/month for 225 credits when you scale. |

| | | |
|---|---|---|
| Question diagrams & content images | Cloudinary (same account) | Included in free tier credits. Images are small and infrequently changed. |
| OCR (convert handwritten photo to text for AI marking) | Google Vision API | Free: 1,000 images/month. Then $1.50 per 1,000 images. At 90 answers/month per learner with ~30% being photo submissions: ~27 OCR calls/learner/month on Free tier. |
| Alternative OCR (self-hosted, free) | Tesseract (open source) | Free. Works well for printed/typed text. Struggles with messy handwriting. Viable for MVP to eliminate Google Vision cost. |

### 8.5 Backend Hosting & Database

| Component | Recommended Service | Cost |
|---|---|---|
| Database & Auth | Supabase | Free: $0/month up to 50,000 monthly active users, 500MB DB, 1GB storage. Pro: $25/month for 100,000 MAUs, 8GB DB, 100GB storage. Includes built-in auth, real-time subscriptions, and REST API. |
| Backend API server | Railway | Hobby: $5/month includes $5 in compute credits (pay-per-second). Pro: $20/month. Sufficient for early production. Scale to dedicated servers when learner count grows. |
| File storage (non-video, non-image) | Supabase Storage (included in plan) | Included in Supabase plan. Use for PDFs, rubric files, lesson materials. |

### 8.6 Per-Learner Monthly Cost Estimate

The following estimates are based on realistic usage patterns per learner tier. Tutor-covered learners (inside a tutor space) are billed against the tutor's plan, not individually. All AI costs use DeepSeek V4 Flash with caching applied to the system prompt. Video delivery assumes 50MB average per lesson stream at 480p (Free) and 150MB at 1080p (Premium). ZAR conversion at R18.40/USD.

| Cost Component | Free Tier | Basic | Standard | Premium |
|---|---|---|---|---|
| AI Marking (DeepSeek) | 90 marks ~$0.02 / R0.37 | 600 marks ~$0.11 / R2.02 | 1,200 marks ~$0.22 / R4.05 | 2,400 marks ~$0.43 / R7.91 |
| Video Delivery (Bunny) | 4.5GB ~$0.05 / R0.92 | 8GB ~$0.08 / R1.47 | 12GB ~$0.12 / R2.21 | 18GB ~$0.18 / R3.31 |
| Video Storage (Bunny) | ~$0.003 / R0.06 | ~$0.003 / R0.06 | ~$0.003 / R0.06 | ~$0.003 / R0.06 |
| Image / OCR | Negligible (free tier) | ~$0.02 / R0.37 | ~$0.03 / R0.55 | ~$0.05 / R0.92 |
| Hosting share | ~$0.05 / R0.92 | ~$0.08 / R1.47 | ~$0.10 / R1.84 | ~$0.15 / R2.76 |
| TOTAL / learner / month | ~$0.12 / R2.21 | ~$0.29 / R5.34 | ~$0.46 / R8.46 | ~$0.84 / R15.46 |

**At R80–120/month for Premium tier, your infrastructure cost is R15.46/learner. That is an 81–87% gross margin on infrastructure alone before any other costs.**

### 8.7 Platform Fixed Costs (Monthly, Both Products)

These costs apply regardless of learner count and represent your minimum monthly spend to keep both platforms running:

| Service | Plan | Monthly Cost (USD) | Monthly Cost (ZAR) |
|---|---|---|---|
| Supabase | Free (up to 50k MAUs) | $0 | R0 |
| Railway | Hobby plan | $5 | R92 |
| Bunny.net | Pay-as-you-go (min spend) | ~$1 | R18 |
| Cloudinary | Free tier | $0 | R0 |
| DeepSeek API | Pay-per-use (no minimum) | $0 | R0 |
| Domain names (x2) | Annual / 12 | ~$2 | R37 |
| SSL certificates | Let's Encrypt (free) | $0 | R0 |
| Total fixed base | | ~$8 | ~R147 |

Your total platform cost at 100 learners on mixed tiers would be approximately R147 (fixed) + R500–800 (variable) = R650–950/month. At 1,000 learners it scales roughly linearly on the variable side while fixed costs barely move until you need to upgrade Supabase or Railway.

### 8.8 Cost Scaling Milestones

| Milestone | Trigger | Action Required | Estimated Cost Increase |
|---|---|---|---|
| 50,000+ MAUs | Supabase free tier limit reached | Upgrade to Supabase Pro: $25/month | +$25/month (R460) |
| High video traffic | Bunny delivery bill exceeds $50/month | Evaluate Bunny volume pricing tier for discounts on bandwidth | Discount applies — cost grows slower |
| Heavy image uploads | Cloudinary free credits exhausted | Upgrade to Cloudinary Plus: $89/month | +$89/month (R1,638) — covers very large scale |
| AI cost spike | DeepSeek bill exceeds $50/month | Review prompt caching implementation. Consider self-hosting DeepSeek-R1 on a VPS for fixed cost. | VPS ~$20–40/month replaces per-token billing |
| POPIA / data residency | School contract requires SA data storage | Switch to self-hosted LLM on SA cloud (AWS Cape Town, Azure SA North) | +$50–200/month depending on model size |

## 9. Classroom Scenario Validation

Before finalising the feature set, a real-world teacher scenario was walked through against the system design to verify that the platform as described would actually support how teachers and learners behave in a live classroom. This is a requirements validation exercise — not a test of the software, but a test of the design.

### 9.1 The Scenario

**"Class, please log onto EduSchool and complete Exercise 8 in the textbook." Learners log in and work through the exercise at their own pace. After 20 minutes, the teacher checks who has logged in, how many questions have been answered, and which learners need a bit more help — without leaving her desk or walking from learner to learner.**

### 9.2 Step-by-Step Walkthrough

| Step | What Happens | Does Our Design Support This? |
|---|---|---|
| Teacher says: open Exercise 8 | Teacher needs to push a specific exercise to the class so it appears immediately on every learner's dashboard. | ■ Gap identified. No assignment push feature existed. Teacher would have had to tell learners to navigate manually — too slow in a live class. → Feature added: Assignment Push (see Section 10). |
| Learners log in and find the task | Learner opens EduSchool and needs to see the exercise immediately without navigating through curriculum → grade → subject → topic. | ■ Gap identified. Without a pinned active task, navigation takes 4 steps and costs several minutes of a 20-minute session. → Resolved by Assignment Push: exercise appears pinned at top of dashboard on login. |
| Learners complete questions | Each learner works through questions at their own pace. AI marks each answer and returns feedback instantly. | ✓ Fully supported. AI marking, step-by-step hints, instant feedback, and self-paced progression all exist in the current design. |
| Learners watch video if stuck | Learner can choose to watch the linked video lesson before or during the exercise. | ✓ Fully supported. Video lessons are linked to topics and available during the exercise. |
| Teacher checks after 20 minutes | Teacher needs to see: who logged in, who started the exercise, scores so far, who has not started. | ■ Gap identified. Class dashboard shows cumulative historical data, not a live session view filtered by the current assignment. → Feature added: Live Session View (see Section 11). |
| Teacher identifies who needs help | Teacher needs to spot struggling learners quickly — at a glance, not by clicking into each learner individually. | ■ Gap identified. Existing flags fire based on historical thresholds, not real-time session performance. → Resolved by Live Session View: colour-coded learner tiles update every 60 seconds. |

| | | |
|---|---|---|
| Teacher acts without leaving desk | Teacher sends a nudge, assigns a remedial lesson, or flags a learner for a follow-up conversation. | ✓ Supported. Nudge, redirect to lesson, and private note all exist in current design. |
| Session ends — teacher reviews | Teacher wants a one-page summary of the session: completion rate, scores, class average, common mistakes. | ■ Gap identified. No per-assignment report existed. → Feature added: Assignment Report (see Section 12). |

### 9.3 Validation Outcome

The scenario walkthrough identified three features missing from the design that are not optional — they are core to the teacher use case working in a real classroom. Importantly, none of these require architectural changes. All the underlying data is already being captured in the current design (login timestamps, answer timestamps, scores, session activity). What was missing was the presentation layer to surface that data in the right way at the right time.

The assumption baked into this scenario — and confirmed as the correct model — is that content exists on the platform before the lesson starts. The teacher's job during class is to push and monitor, not to create. Content creation is a preparation activity done outside of class time.

| Feature Gap | Resolution | Section |
|---|---|---|
| No way to push a specific exercise to a class instantly | Assignment Push feature added | Section 10 |
| No live session view filtered by the current assignment | Live Session View feature added | Section 11 |
| No per-assignment summary report after the session | Assignment Report feature added | Section 12 |

## 10. Assignment Push (EduSchool)

Assignment Push allows a teacher to select any existing topic, exercise, or question set from the platform and push it to their class as the active task. The content must already exist on the platform before the lesson starts. The teacher's job during class is to push and monitor, not to create.

**The teacher does not send a link. Does not write on the board. Does not wait for learners to navigate. They push the exercise and it appears at the top of every learner's dashboard immediately.**

### 10.1 Teacher: Pushing an Assignment

1. Teacher opens EduSchool and navigates to their class dashboard.
2. Teacher clicks "Push Assignment."
3. A content browser opens showing all topics and exercises available to this class (content the teacher created, school-wide content, or platform-wide shared content).
4. Teacher selects the exercise (e.g. Exercise 8 — Trigonometry Practice).

5. Teacher optionally sets a session duration (e.g. 20 minutes). This starts a countdown visible on the teacher's Live Session View.
6. Teacher clicks "Push to Class."
7. Every learner currently logged in sees the assignment appear pinned at the top of their dashboard immediately. Learners who log in during the session see it on arrival.

### 10.2 Learner: Receiving a Pushed Assignment

1. Learner logs into EduSchool.
2. At the top of the dashboard, a highlighted banner appears: "Active Task: Exercise 8 — Trigonometry Practice (set by Ms Dlamini)."
3. Learner taps the banner and is taken directly into the exercise. No navigation through curriculum or grade menus required.
4. Learner works through questions at their own pace. AI marks each answer and returns feedback instantly.
5. If a learner finishes early, they can continue with other content in the topic or move to the next lesson — the active task banner remains until the teacher ends the session.

### 10.3 Assignment States

| State | Description | Visible To |
|---|---|---|
| Active | Currently pushed to the class. Pinned on all learner dashboards. Live session view running. | Teacher + all learners in class |
| Ended | Teacher has closed the session or duration has elapsed. Pinned banner removed from learner dashboards. Data locked for report generation. | Teacher (in assignment history) |
| Scheduled | Teacher has set a future push time (e.g. for tomorrow's lesson). | Teacher only |
| Draft | Teacher has selected content but not yet pushed. | Teacher only |

### 10.4 Rules

- Only one assignment can be active per class at a time. Pushing a new one ends the current session.
- A teacher can push any content available to their class: content they created, school-wide content, or platform-wide shared content they have access to.
- Content must exist on the platform before it can be pushed. The push feature is not a content creation tool.
- Learners who are not logged in when the assignment is pushed will see it pinned when they next log in, as long as the session is still active.
- The teacher can end a session early at any time. This generates the Assignment Report immediately.

## 11. Live Session View (EduSchool)

The Live Session View is a real-time panel the teacher opens during an active assignment. It shows the entire class at a glance — who is in, who is progressing, who is struggling, and who has not started — updated every 60 seconds. The teacher scans the room digitally instead of walking from desk to desk.

**The teacher does not need to click into individual learner profiles. The session view shows the whole class on one screen. Problems are visible in seconds.**

## 11.1 Learner Tiles

Each learner in the class is represented by a tile. Tiles are colour-coded and update every 60 seconds based on activity and performance during the current session:

| Tile Colour | Meaning | What the Teacher Sees |
|---|---|---|
| Green | Active and doing well. | Learner has logged in, started the assignment, and is scoring above the performance threshold (configurable, default 60%). |
| Amber | Active but struggling. | Learner has logged in and started, but is scoring below the threshold or has failed the same question more than twice. |
| Red | Not started or inactive. | Learner has logged in but has not opened the assignment, or has been inactive for more than 5 minutes. |
| Grey | Not logged in. | Learner has not logged into EduSchool during this session. |

## 11.2 Tile Detail

Each tile shows the learner's name and three numbers at a glance:

- Questions answered out of total in the assignment (e.g. 4/10).
- Current score percentage (e.g. 75%).
- Time since last activity (e.g. 2 min ago).

The teacher taps any tile to expand it and see which specific questions the learner answered correctly or incorrectly during this session, without leaving the session view.

## 11.3 Teacher Actions from the Session View

Without leaving the session view, the teacher can:

| Action | How | Result |
|---|---|---|
| Send a nudge | Tap learner tile → tap "Send Nudge." | Learner receives a notification: "Ms Dlamini says: please start the exercise." |
| Redirect to a lesson | Tap learner tile → tap "Assign Lesson." Select a video lesson. | A specific lesson appears pinned on that learner's dashboard above the active assignment. |
| Add a private note | Tap learner tile → tap "Add Note." | Note saved to learner's profile. Visible to teacher and school admin only. |

| | | |
|---|---|---|
| Flag for follow-up | Tap learner tile → tap “Flag.” | Learner appears in the “Needs Follow-Up” list on the teacher’s dashboard after the session ends. |
| Notify parent | Tap learner tile → tap “Notify Parent.” | Parent portal shows a notification: “Your child may need extra support with this topic.” |

### 11.4 Class Summary Bar

At the top of the session view, a summary bar shows the class totals at a glance:

- Total logged in / total in class (e.g. 32/35).
- Total who have started the assignment (e.g. 28/35).
- Class average score so far (e.g. 64%).
- Number of amber tiles (struggling) and red/grey tiles (not started).
- Session timer countdown if a duration was set.

### 11.5 Refresh Rate & Performance

- Tiles refresh every 60 seconds automatically. Teacher can also tap “Refresh Now.”
- The session view is optimised for tablets and large phone screens. Teacher can keep it open on their device while moving around the room if needed.
- The session view does not require a page reload. It uses real-time data from the platform’s activity feed.

## 12. Assignment Report (EduSchool)

When a session ends — either by the teacher closing it or the duration elapsing — the platform automatically generates an Assignment Report. This is a one-page summary of what happened during the session. The teacher has this in hand before the next class begins.

**The teacher does not mark a single script. Does not tally scores on a spreadsheet. Does not wait until the end of term to find out who is struggling. The report is ready the moment the session ends.**

### 12.1 Report Contents

| Section | Content |
|---|---|
| Session Summary | Assignment name, date, class, duration, total learners in class, total who logged in, total who started the assignment. |
| Completion Overview | A simple bar: X learners completed all questions, Y completed some, Z did not start. Percentage completion across the class. |
| Class Score Summary | Class average score. Highest and lowest scores. Score distribution (how many scored 0–40%, 40–60%, 60–80%, 80–100%). |

| | |
|---|---|
| Per-Learner Table | One row per learner: name, questions answered, score, time spent, status (completed / in progress / not started). Sortable by score. |
| Common Mistakes (AI-generated) | The AI aggregates all incorrect answers across the class for each question and produces a plain-language summary: “18 out of 32 learners made an error on Question 3 — most confused the sine and cosine rules when the triangle was obtuse.” This tells the teacher exactly what to address in the next lesson. |
| Needs Follow-Up | Learners flagged during the session, plus any learner who scored below the threshold. One-tap action to notify parents or assign remedial content from the report. |

### 12.2 Report Delivery

- Available immediately in the teacher’s dashboard under “Assignment History.”
- Optional email summary sent to the teacher at session end (toggled in notification settings).
- School Admin can view all assignment reports across all classes in the school.
- The per-learner table can be exported as a CSV for use in the school’s existing reporting system.

### 12.3 Parent Visibility

If a learner’s parent has a linked parent portal account, the parent sees a simplified version of their child’s assignment result after the session ends:

- Assignment name and date.
- Questions answered and score received.
- Whether the teacher flagged the learner for follow-up.
- AI feedback summary for any questions the learner got wrong.

The parent does not see other learners’ scores or the class-wide report. They see only their child.

## 13. Validated Scenario Re-Walkthrough

With the three new features in place, the classroom scenario now plays out as follows:

| Step | What Happens | Feature Used |
|---|---|---|
| Teacher says: open Exercise 8 | Teacher opens class dashboard, taps Push Assignment, selects Exercise 8 from the content browser, sets 20-minute timer, taps Push to Class. | Assignment Push (Section 10) |
| Learners log in | Every learner who logs in sees Exercise 8 pinned at the top of their dashboard. One tap to start. No navigation required. | Assignment Push — active task banner |
| Learners work through questions | Each learner answers at their own pace. AI marks instantly. Feedback returned in seconds. Hints available if stuck. | AI Marking Engine (Section 4, Shared Core) |
| Learners watch video if stuck | Video lesson linked to the topic is available during the exercise. | Video Delivery (Bunny Stream) |

| | | |
|---|---|---|
| Teacher checks at 20 minutes | Teacher opens Live Session View. Sees 32/35 logged in, 28 started, class average 61%, 6 amber tiles, 3 grey tiles. | Live Session View (Section 11) |
| Teacher identifies who needs help | Teacher sees 6 amber tiles at a glance. Taps each to see which questions they struggled on. No clicking through individual profiles. | Live Session View — colour-coded tiles |
| Teacher acts | Teacher sends a nudge to 3 grey-tile learners. Assigns a remedial video to 2 amber learners. Flags 1 learner for parent notification. | Live Session View — in-session actions |
| Session ends | Teacher taps End Session. Assignment Report generated immediately. | Assignment Report (Section 12) |
| Teacher reviews report | Report shows: 28/35 completed, class average 64%, common mistake on Question 3 identified by AI across 18 learners. Teacher knows exactly what to address next lesson. | Assignment Report — common mistakes (AI) |
| Parents informed | Parents with portal access see their child's score, AI feedback, and any follow-up flag — before the school day ends. | Parent Portal + Assignment Report |

### 13.1 Final Validation

| Original Question | Answer |
|---|---|
| Will the teacher be able to push a task without learners navigating manually? | ✓ Yes. Assignment Push pins the exercise to every learner's dashboard in one action. |
| Will learners be able to find and start the exercise quickly? | ✓ Yes. One tap from the dashboard to the exercise. No navigation steps. |
| Will the teacher be able to see who has logged in and who has answered questions after 20 minutes? | ✓ Yes. Live Session View shows this at a glance, colour-coded, updated every 60 seconds. |
| Will the teacher be able to identify who needs help without leaving their desk? | ✓ Yes. Amber and grey tiles flag struggling and absent learners immediately. |
| Will the teacher be able to act on a struggling learner without leaving their desk? | ✓ Yes. Nudge, redirect to lesson, flag, and parent notify all available from the tile. |
| Will the teacher get a report without marking anything? | ✓ Yes. Assignment Report generated automatically at session end. AI identifies common mistakes across the class. |
| Will learners learn at their own pace and get feedback without depending on the teacher? | ✓ Yes. AI marking is instant. Hints are self-serve. Learners never wait for the teacher to mark. |
| Will parents know how their child performed? | ✓ Yes. Parent portal updated automatically at session end. |

# 14. EduTutor: Platform Modes

EduTutor supports two operating modes on the same platform. The mode is set per learner account and can be changed at any time. Both modes share the same AI marking engine, content delivery, and progress tracking.

| Mode | Who Is Involved | Who Monitors | Best For |
|---|---|---|---|
| Learner-Parent Mode | Learner + Parent only. No tutor assigned. | Parent monitors via portal. Notifications alert parent when learner is inactive, struggling, or completes a topic. | Families who want self-managed learning with parental oversight. No tutor cost. Parent is the accountability partner. |
| Tutor-Learner Mode | Tutor + Learner. Parent can be optionally linked. | Tutor monitors via dashboard. Alerts go to tutor and learner. Parent optionally linked for visibility. | Learners who need structured guidance. Tutor plans sessions using live data. Parent stays informed without managing. |

A learner can start on Learner-Parent Mode and add a tutor later. A tutor can invite a learner who has no existing account. The mode switch preserves all progress history.

# 15. EduTutor Scenario Validation

Two real-world scenarios were walked through against the EduTutor design to validate that the platform supports how tutors, learners, and parents actually behave. Gaps were identified and resolved as new features.

## 15.1 Scenario A: Scheduled Session at 6pm

**“A learner has a session with their tutor at 6pm. Both need to be reminded. Both confirm they are attending. The session happens. The tutor uses the learner’s data to guide what they work on. After the session, both know what was covered and what to do before the next one.”**

| Step | What Happens | Does Our Design Support This? |
|---|---|---|
| Session is scheduled | Tutor creates a session in EduTutor: selects the learner, sets date and time (6pm), adds an optional note (e.g. ‘focus on quadratic equations’). | ■ Gap. No session scheduling feature existed. → Feature added: Session Scheduling (Section 16). |
| Reminder alerts sent | At 5:45pm, both tutor and learner receive a notification: “Your session starts in 15 minutes.” At 6:00pm, another alert fires. | ■ Gap. No notification/alert system existed in the design. → Feature added: Notification & Alert System (Section 18). |

| | | |
|---|---|---|
| Attendance confirmed | Tutor and learner each tap “Confirm Attendance” in the app. Session is marked as confirmed. If either does not confirm by 6:05pm, the other is notified. | ■ Gap. No attendance confirmation existed. → Feature added: Attendance Confirmation (Section 16). |
| Session prep | Before the session, tutor opens the session prep view: sees what the learner struggled with since the last session, which topics were attempted, AI feedback received. | ✓ Supported. Tutor dashboard already captures this. Session prep view surfaces it per learner per session. |
| Session happens | Phase 1: session happens externally (Zoom, in person). Phase 2: session happens inside EduTutor via integrated video, whiteboard, and chat. | ■ Phase 1 supported by design (platform tracks session, not host it). Phase 2 → Feature added: In-Session Tools (Section 17). |
| Tutor assigns post-session work | After the session, tutor assigns specific exercises or topics for the learner to complete before the next session. | ✓ Supported via Assignment Push (adapted for individual learner, not class). See Section 10. |
| Learner gets summary | Learner sees a session summary in their dashboard: what was covered, what was assigned, when the next session is. | ■ Gap. No session summary existed. → Resolved by Session Report (Section 16). |
| Parent is informed | Parent receives a notification that the session took place and can see the session summary in the parent portal. | ■ Gap. Parent alerts did not exist for EduTutor. → Resolved by Notification System + parent linking in Tutor-Learner Mode. |

## 15.2 Scenario B: Learner-Parent Mode — Self-Managed Learning

**“A parent subscribes for their child. No tutor is assigned. The child studies independently. The parent wants to know if the child is actually logging in, making progress, and not just sitting on the app doing nothing. If the child has not logged in for two days, the parent wants to know.”**

| Step | What Happens | Does Our Design Support This? |
|---|---|---|
| Parent subscribes | Parent creates an account, selects Learner-Parent Mode, sets up the learner’s profile (name, grade, curriculum). | ✓ Supported. Registration and onboarding exist. Learner-Parent Mode is a new mode flag added in Section 14. |
| Parent sets learning goals | Parent sets a daily target: e.g. at least 1 lesson and 5 questions per day. | ■ Gap. No daily goal setting existed. → Feature added: Learning Goals (Section 19). |

| | | |
|---|---|---|
| Learner logs in and studies | Child logs in, works through lessons and questions at their own pace. AI marks and gives feedback. | ✓ Fully supported. Core platform. |
| Parent receives daily summary | Each evening, parent receives a notification: “Your child completed 2 lessons and answered 8 questions today. Score average: 72%.” | ■ Gap. No daily parent summary notification existed. → Resolved by Notification System (Section 18). |
| Child misses two days | Child does not log in for 48 hours. Parent receives an alert: “Your child has not been active on EduTutor for 2 days.” | ■ Gap. No inactivity alert existed. → Resolved by Notification System — inactivity trigger (Section 18). |
| Child is struggling | Child fails consolidation on the same topic twice. Parent is alerted: “Your child may need extra support with Trigonometry.” | ■ Gap. Struggle alerts went to tutor only. → Extended to parent in Learner-Parent Mode (Section 18). |
| Parent reviews progress | Parent opens the parent portal and sees full activity history, scores, AI feedback, and which topics are complete. | ✓ Supported. Parent portal exists in the design. Extended to EduTutor in this mode. |

# 16. Session Scheduling & Attendance (EduTutor)

Session Scheduling allows tutors to plan sessions with individual learners inside EduTutor. The platform manages reminders, attendance confirmation, session prep, and post-session summaries. The actual lesson can happen anywhere — in person, Zoom, Teams, or (in Phase 2) inside EduTutor’s own session room.

## 16.1 Creating a Session

1. Tutor navigates to “My Sessions” and taps “Schedule Session.”
2. Tutor selects the learner, date, time, and estimated duration.
3. Tutor optionally adds a focus note: e.g. ‘Quadratic equations — learner struggled with factorisation this week.’
4. Tutor optionally pre-assigns exercises for the learner to attempt before the session.
5. Session is saved. Learner receives an immediate notification: “Your tutor has scheduled a session for [date] at [time].”
6. If a parent is linked in Tutor-Learner Mode, they also receive the session notification.

## 16.2 Attendance Confirmation

Attendance confirmation closes the loop on whether the session will actually happen. Both parties must confirm. If either does not confirm, the other is alerted.

| Time | Event |
|---|---|

| | |
|---|---|
| 24 hours before | Reminder notification sent to tutor and learner: “Session tomorrow at 6pm. Tap to confirm attendance.” |
| 1 hour before | Second reminder sent if either party has not confirmed. |
| 15 minutes before | Final reminder sent to both. Session prep view opens automatically for the tutor. |
| Session time | Both receive a “Your session starts now” notification. |
| 5 minutes after | If learner has not confirmed attendance or entered the session room (Phase 2), tutor is notified: “Learner has not joined yet.” |
| Either cancels | Other party is notified immediately. Session is marked as cancelled. Reschedule prompt appears. |

### 16.3 Session Prep View

When the session time approaches, the tutor’s app surfaces a session prep card automatically. This gives the tutor everything they need to run a focused session without digging through the dashboard:

- Topics the learner attempted since the last session.
- Questions they got wrong — with the AI’s error explanation.
- Consolidation exercises passed or failed.
- Pre-session exercises assigned (if any) and whether the learner completed them.
- Suggested focus area based on performance data: e.g. “Learner struggled most with factorisation — 3 incorrect attempts this week.”

### 16.4 Post-Session Summary

After the session, the tutor logs what was covered in a quick post-session form (this takes under 2 minutes):

1. Topics covered during the session (selected from a list or typed).
2. Overall session rating: productive / needs follow-up / learner struggled.
3. Post-session exercises assigned to the learner before the next session.
4. Next session date (optional — can be scheduled directly from this screen).

The platform generates a Session Summary that both the learner and (if linked) the parent can see:

- Date, duration, tutor name.
- Topics covered.
- Exercises assigned before next session.
- Next session date and time.
- Tutor’s overall rating (visible to parent, not shown as a number to learner — shown as “Session complete” or “Extra practice recommended”).

## 17. In-Session Tools: Video, Whiteboard & Chat

In-session tools bring the lesson experience inside EduTutor so the platform owns the full learning loop — from scheduling through to the live explanation and back to the follow-up exercises. Without this, the session data (what was explained, what was drawn, what was said) is lost the moment the learner closes Zoom.

**Phase 1: Sessions are scheduled and tracked in EduTutor. The actual call happens externally. Phase 2: Video, whiteboard, and chat are built into EduTutor. The session stays inside the platform. Everything is recorded and linked to the learner's topic.**

## 17.1 Phase 1 — External Session (MVP)

In Phase 1, EduTutor manages everything around the session but does not host it:

- Scheduling, reminders, and attendance confirmation are fully functional.
- Session prep view and post-session summary are fully functional.
- The session card shows a "Join Session" button that opens the tutor's preferred external tool (Zoom, Teams, Google Meet) via a link the tutor pastes when scheduling.
- The platform records that the session happened, its duration (manually entered by tutor), and what was covered.

This is sufficient for MVP. It validates the scheduling and monitoring workflow before the more complex in-session build.

## 17.2 Phase 2 — In-Platform Session Room

In Phase 2, a Session Room is built inside EduTutor using third-party SDKs embedded into the platform. The tutor and learner never need to leave the app.

| Tool | Technology | What It Does |
|---|---|---|
| Video Call | Daily.co or LiveKit SDK (embedded) | Peer-to-peer video between tutor and learner. One tap to start from the session card. Works on mobile and desktop. No account required for learner. |
| Collaborative Whiteboard | Tldraw (open source, embeddable) | Shared canvas. Tutor draws, writes, and explains in real time. Learner can draw too. Any question from the platform can be imported directly onto the whiteboard — tutor explains the exact question the learner got wrong. |
| Chat | Built-in (simple text) | Text chat during the session. Useful when video connection is weak. Tutor can share formulas, links, or quick notes. Chat log saved to session record. |
| Session Recording | Daily.co / LiveKit cloud recording | Session is recorded automatically (with consent). Recording stored against the learner's profile, linked to the topic covered. Learner can replay the explanation later. |

## 17.3 The Whiteboard Differentiator

The whiteboard is the most important in-session feature because it connects the live explanation directly to the platform's learning data:

- Tutor opens a question the learner got wrong in their exercise history.
- With one tap, that question is imported onto the shared whiteboard.

- Tutor draws the correct working, step by step, explaining each line.
- The whiteboard snapshot is saved at session end, linked to that question, and stored in the learner's profile.
- If the tutor chooses, the whiteboard recording can be converted into a video lesson asset available to other learners on the topic.

No other platform in the SA market connects a live tutoring explanation directly to the specific question a learner failed. This is a genuine differentiator.

### 17.4 Phase 2 Technology Costs

| Service | Cost |
|---|---|
| Daily.co | Free up to 10,000 minutes/month. Then $0.00099/minute per participant. A 60-minute session with 2 participants = ~$0.12 / R2.21. |
| LiveKit | Open source. Self-hostable for zero variable cost. Requires a server (~$20/month). Recommended once session volume justifies it. |
| Tldraw | Open source. Free to embed. No per-user cost. |
| Recording storage | Stored on Bunny.net. Same cost as lesson videos: ~$0.005/GB/month. |

## 18. Notification & Alert System

Notifications are the nervous system of both EduTutor and EduSchool. The accountability loop only works if the right person is nudged at the right moment. Every notification is configurable — each user chooses which alerts they receive and how (push notification, SMS, email). All defaults are on.

### 18.1 Learner Notifications

| Trigger | Notification | Mode |
|---|---|---|
| Session in 15 minutes | "Your session with [Tutor] starts in 15 minutes. Tap to confirm attendance." | Tutor-Learner |
| Session starting now | "Your session is starting now." | Tutor-Learner |
| Assignment pushed | "[Teacher/Tutor] has set a new task: [Exercise name]. Open EduTutor to start." | Both |
| AI feedback ready | "Your answer has been marked. Tap to see your feedback." | Both |
| Consolidation passed | "Well done! You've passed [Topic]. The next topic is now unlocked." | Both |
| Consolidation failed | "Keep going — review the lesson on [Topic] and try again." | Both |
| Post-session work assigned | "[Tutor] has assigned exercises for you to complete before your next session." | Tutor-Learner |

| | | |
|---|---|---|
| Daily reminder (if inactive) | If learner has not logged in by a configurable time (e.g. 5pm): “Don’t forget to study today.” | Both |

### 18.2 Tutor Notifications

| Trigger | Notification |
|---|---|
| Session in 15 minutes | “Your session with [Learner] starts in 15 minutes. Session prep is ready.” |
| Learner confirms attendance | “[Learner] has confirmed attendance for tonight’s session.” |
| Learner has not confirmed | “[Learner] has not confirmed attendance for tonight’s session (starts in 1 hour).” |
| Learner struggling flag | “[Learner] has failed consolidation on [Topic] twice this week. Consider addressing this in your next session.” |
| Learner inactive 48 hours | “[Learner] has not been active on EduTutor for 2 days.” |
| Lesson request submitted | “[Learner] has requested a lesson on [Topic].” |
| Package limit approaching | “You are at 80% of your learner limit. Consider upgrading your package.” |

### 18.3 Parent Notifications

| Trigger | Notification | Mode |
|---|---|---|
| Session scheduled | “[Tutor] has scheduled a session for [Learner] on [date] at [time].” | Tutor-Learner |
| Session completed | “[Learner]’s session with [Tutor] is complete. Tap to view the session summary.” | Tutor-Learner |
| Daily activity summary | “[Learner] completed [X] lessons and answered [Y] questions today. Average score: [Z]%.” | Both |
| Learner inactive 48 hours | “[Learner] has not logged into EduTutor for 2 days.” | Both |
| Learner struggling | “[Learner] may need extra support with [Topic]. They have failed consolidation twice.” | Both |
| Topic completed | “[Learner] has completed [Topic]. Great progress!” | Both |
| Assignment result | “[Learner] completed [Exercise] and scored [X]%. Tap to view their feedback.” | Both |
| Tutor flag | “[Tutor] has flagged [Learner] for follow-up after today’s session.” | Tutor-Learner |

### 18.4 Notification Delivery Channels

| Channel | Default | Notes |
|---|---|---|
| Push notification (in-app) | On | Requires EduTutor app installed. Instant delivery. |

| | | |
|---|---|---|
| Email | On | Fallback if push is not available. Daily digest option for parents. |
| SMS | Off | Optional. User provides mobile number. Charged per SMS. Recommended for session reminders only. |

All notification types are individually toggleable per user from their profile settings. Tutors can also set quiet hours (e.g. no notifications after 9pm). Parents can choose daily digest instead of individual notifications to reduce noise.

# 19. Learning Goals (Learner-Parent Mode)

In Learner-Parent Mode, the parent sets a daily or weekly learning goal for their child. This creates a measurable target that drives the notification system and progress tracking.

| Goal Setting Option | Example |
|---|---|
| Daily lessons | At least 1 video lesson per day. |
| Daily questions | At least 5 questions answered per day. |
| Weekly topics | Complete at least 1 full topic per week. |
| Daily study time | At least 20 minutes of active platform use per day. |

- Goals are set by the parent from the parent portal.
- The learner sees their daily goal progress on their dashboard (e.g. a simple progress bar: 3/5 questions done today).
- If the learner has not met their goal by a configurable time (e.g. 7pm), the parent receives an alert.
- Weekly goal summary sent to parent every Sunday evening.
- Goals are suggestions — the platform does not block a learner from doing more or less. It tracks and reports.

# 20. Validated Tutor & Parent Scenario Re-Walkthrough

## 20.1 Scenario A: 6pm Tutor Session — Replayed

| Step | What Happens Now | Feature |
|---|---|---|
| Tutor schedules session | Tutor creates session in EduTutor: learner, 6pm, 60 min, focus note added. | Session Scheduling (S16) |
| Learner notified | Learner receives immediate notification: “Session scheduled for tonight at 6pm.” | Notification System (S18) |
| Parent notified (if linked) | Parent receives: “A session has been scheduled for [Learner] tonight at 6pm.” | Notification System (S18) |
| 5:45pm — reminder fires | Both tutor and learner receive: “Session in 15 minutes. Tap to confirm attendance.” | Notification System (S18) |

| | | |
|---|---|---|
| Both confirm attendance | Tutor taps Confirm. Learner taps Confirm. Both see: “Attendance confirmed.” | Session Scheduling (S16) |
| Tutor reviews session prep | Session prep card opens: learner failed factorisation twice this week. Tutor knows exactly what to focus on before saying hello. | Session Scheduling (S16) |
| Session happens (Phase 1) | Tutor taps Join Session — opens Zoom link. Lesson takes place externally. | Phase 1 external (S17) |
| Session happens (Phase 2) | Tutor taps Start Session Room. Video, whiteboard, and chat open inside EduTutor. Tutor imports the failed question onto the whiteboard and explains it live. | In-Session Tools (S17) |
| Post-session logged | Tutor fills in 2-minute post-session form: topics covered, exercises assigned, next session date. | Session Scheduling (S16) |
| Learner sees summary | Learner’s dashboard shows session summary: covered, assigned exercises, next session. | Session Scheduling (S16) |
| Parent informed | Parent receives: “Session complete. Tap to view summary.” | Notification System (S18) |

## 20.2 Scenario B: Learner-Parent Mode — Replayed

| Step | What Happens Now | Feature |
|---|---|---|
| Parent subscribes | Parent registers, selects Learner-Parent Mode, sets up child’s profile. | Platform Modes (S14) |
| Parent sets daily goal | Parent sets: 1 lesson + 5 questions per day. | Learning Goals (S19) |
| Child logs in and studies | Child sees daily goal progress bar on dashboard. Studies at own pace. AI marks answers instantly. | Core platform + Learning Goals |
| Evening summary to parent | 7pm: Parent receives “[Child] completed 2 lessons and 8 questions today. Average score: 72%.” | Notification System (S18) |
| Child misses two days | 48 hours of inactivity triggers: “[Child] has not logged in for 2 days.” Parent can act. | Notification System (S18) — inactivity trigger |
| Child fails consolidation twice | Parent receives: “[Child] may need extra support with Trigonometry.” | Notification System (S18) — struggle trigger |
| Parent reviews portal | Full activity history, scores, AI feedback, topic progress all visible. | Parent Portal (S7 / EduTutor extended) |

| | | |
|---|---|---|
| Parent decides to add a tutor | Parent switches mode to Tutor-Learner. Tutor invited by email. All progress history preserved. | Platform Modes (S14) |

### 20.3 Final Validation

| Question | Answer |
|---|---|
| Will tutor and learner both be alerted before the session? | ✓ Yes. 24h, 1h, and 15min reminders via push/email/SMS. |
| Can both confirm attendance? | ✓ Yes. Confirmation tap in the app. Non-confirmation triggers an alert to the other party. |
| Will the tutor know what to focus on before the session starts? | ✓ Yes. Session prep card surfaces the learner's struggle data automatically. |
| Can the session happen inside EduTutor? | ✓ Phase 2: Yes. Video, whiteboard, chat, and recording inside the platform. |
| Will the whiteboard connect to the learner's actual failed questions? | ✓ Yes. One-tap import of any question from the learner's exercise history onto the whiteboard. |
| Will the parent know the session happened and what was covered? | ✓ Yes. Session summary notification and portal update immediately after session ends. |
| In Learner-Parent Mode, will the parent know if the child is inactive? | ✓ Yes. 48-hour inactivity alert. Daily summary. Struggle flag. Weekly goal report. |
| Can a learner start without a tutor and add one later? | ✓ Yes. Mode switch preserves all progress history. |

# 21. Flashcard System

Flashcards are a first-class learning feature on both EduTutor and EduSchool. They serve two purposes: reinforcing what a learner just studied, and giving teachers, tutors, and parents a quick way to push targeted revision directly to a learner's deck. Every card is linked to a topic so learners always find their cards in context.

| Who Creates | How | Who Sees It |
| --- | --- | --- |
| Learner | Prompted by a pop-up while studying. Types front and back, or accepts a pre-filled suggestion from the platform. | Private to the learner only. Saved under the topic it was created from. |
| Teacher | Creates manually or uses AI generation. Pushes to a class or individual learner. | Pushed learners see the card set in their deck under the relevant topic. |
| Tutor | Creates manually or uses AI generation. Pushes to one or more of their assigned learners. | Pushed learners see the card set in their deck under the relevant topic. |
| Parent | Creates manually or uses AI generation. Pushed to their child only. | Child sees the card set in their deck under the relevant topic. |

## 21.1 Learner-Created Flashcards — The Prompt System

Learners are nudged to create flashcards at natural break points during their study session. The platform does not interrupt mid-question — it prompts at moments when the learner has just absorbed something worth remembering.

### 21.1.1 When the Prompt Appears

| Trigger | Prompt Message |
| --- | --- |
| Learner completes a video lesson | "Want to save a key idea from this lesson as a flashcard? It only takes 10 seconds." |
| Learner answers a question correctly after previously getting it wrong | "You got that one right this time! Want to save it as a flashcard to help it stick?" |
| Learner passes a consolidation exercise | "Well done on completing [Topic]! Create a flashcard to lock in what you learned." |
| Learner opens a topic for the first time | No prompt — learner is still orienting. Prompt fires only after engagement. |

The prompt is a small non-blocking pop-up. The learner can dismiss it with one tap. It does not pause the lesson or require any action. Prompts are limited to a maximum of two per study session to avoid fatigue.

### 21.1.2 Creating the Card

1. Learner taps “Create Flashcard” on the prompt.
2. A card editor opens with two fields: Front (the question or concept) and Back (the answer or explanation).
3. If the prompt was triggered by a lesson or question, the platform pre-fills a suggestion based on the content. Learner can edit or accept.
4. Learner taps Save. Card is saved privately under the current topic.
5. Card appears in the learner’s Flashcard Deck under that topic. It is never visible to anyone else unless the learner chooses to share it (future feature).

### 21.1.3 Where Cards Are Stored

- Each card is tagged to the topic it was created under (e.g. Trigonometry — Grade 11 — CAPS Maths).
- Learner accesses all their cards from “My Flashcards” on their dashboard, organised by subject and topic.
- Cards are also accessible from within the topic page, alongside lessons and exercises.
- Cards created by the learner are private by default and do not appear on any teacher, tutor, or parent view.

## 21.2 Teacher / Tutor / Parent-Created Flashcards

### 21.2.1 Manual Creation

1. Creator navigates to the relevant topic (or to “My Flashcards” from their dashboard).
2. Taps “Create Card Set.” Enters a name for the set (e.g. “Trigonometry Key Formulas”).
3. Adds cards one by one: Front field (question/concept) and Back field (answer/explanation).
4. Reviews the full set before publishing.
5. Pushes to: a specific learner, a group of learners, or the whole class (teacher only).

### 21.2.2 AI-Generated Card Sets

AI generation dramatically reduces the time to create a useful card set. The creator provides a source and the AI produces a ready-to-review set of cards.

| AI Generation Source | How It Works |
|---|---|
| Select a topic from the platform | Creator selects an existing topic (e.g. Organic Chemistry — Grade 12). AI reads the lesson content and question bank for that topic and generates 10–20 flashcard pairs covering the key concepts, definitions, and formulas. |
| Paste text | Creator pastes any text — a chapter summary, a set of notes, a rubric, a past paper memo. AI extracts the key concepts and generates card pairs. |
| Type a prompt | Creator types: e.g. “Generate 10 flashcards on the causes of the Anglo-Boer War at Grade 10 level.” AI generates the set directly. |

1. Creator selects source and taps “Generate.”
2. AI returns a draft set of cards. Creator sees all cards before they go anywhere.

3. Creator can edit, delete, or add cards to the AI draft.
4. Creator approves the set and pushes to learner(s).

### 21.2.3 Pushing Cards to Learners

Once a card set is approved, the creator pushes it:

- Teacher: push to the whole class, a group, or a specific learner. Cards appear in the learner's deck immediately under the linked topic.
- Tutor: push to one or more of their assigned learners. Card set appears in the learner's deck under the linked topic.
- Parent: push to their child only. Card set appears in the child's deck under the linked topic.
- The learner receives a notification: "[Teacher/Tutor/Parent] has added a new flashcard set to your deck: [Set name]. Tap to start studying."
- Pushed cards are clearly labelled in the learner's deck with the creator's name (e.g. "Created by Ms Dlamini") so the learner knows where they came from.

## 21.3 Study Mode: Flip, Rate & Repeat

Study mode is how learners engage with their flashcard decks. It is designed around active recall — the most effective revision technique. The learner tests themselves, self-rates honestly, and the platform uses incorrect cards to build a repeat curve that brings weak cards back more often.

### 21.3.1 Starting a Study Session

1. Learner opens a card set from their deck or from within a topic page.
2. Learner taps "Study Now."
3. Session begins. Cards are shown in a shuffled order by default.

### 21.3.2 The Flip & Rate Loop

| Step | What Happens |
|---|---|
| Card Front shown | The question or concept is displayed. Learner reads it and tries to recall the answer mentally (or says it aloud). |
| Learner taps Flip | Card flips to reveal the Back (answer/explanation). Learner compares their recalled answer to what is shown. |
| Learner self-rates | Two buttons appear: "Got it ✓" and "Missed it ✗." Learner taps honestly based on whether their recalled answer matched. No typing required — this is self-assessed. |
| Card is filed | "Got it" cards move to the Correct pile for this session. "Missed it" cards go into the Repeat Queue and will appear again before the session ends. |
| Next card | Next card in the deck is shown. Process repeats. |

### 21.3.3 The Repeat Curve

The repeat curve ensures that cards the learner struggled with are not forgotten after one session. It operates on two levels:

| Level | Behaviour |
|---|---|
| Within a session | Cards marked “Missed it” are shuffled back into the remaining deck and shown again before the session ends. A card is only removed from the active deck once the learner marks it “Got it”. |
| Across sessions | Cards marked “Missed it” in a session are flagged for priority review. The next time the learner opens that card set, flagged cards appear first. A card that has been marked “Got it” three sessions in a row is marked as Mastered and moves to the back of the rotation. |

### 21.3.4 Session End Summary

When all cards have been seen and the repeat queue is clear, the session ends automatically. The learner sees:

- Total cards in the set.
- Cards correct on first attempt (e.g. 12/18).
- Cards that needed repetition (e.g. 6 cards required 2+ attempts).
- Cards now marked as Mastered vs still in progress.
- A simple performance indicator: e.g. “Strong session — keep it up” or “Focus area: [concept] — review the lesson on this topic.”

### 21.3.5 Card Set Rating

After completing a study session on a pushed card set (one created by a teacher, tutor, or parent), the learner is prompted to rate the set:

- A simple 1–5 star rating or thumbs up/down.
- Optional one-line comment: “These cards helped a lot” or “Some of these were confusing.”
- Ratings are visible to the creator of the card set (teacher, tutor, parent). Not visible to other learners.
- Teacher can see aggregate ratings across their whole class for a given card set — useful for knowing which sets to keep and which to improve.

## 21.4 Flashcard Dashboard Views

### 21.4.1 Learner View

| Section | Content |
|---|---|
| My Cards | All cards the learner created themselves, organised by subject and topic. |
| Pushed to Me | Card sets pushed by teachers, tutors, or parents. Labelled with creator name. |
| Study Progress | Per card set: total cards, mastered count, in-progress count, last studied date. |
| Flagged Cards | Cards marked “Missed it” in the last session, ready for priority review. |

### 21.4.2 Teacher / Tutor View

| Section | Content |
|---|---|
| My Card Sets | All sets created by this teacher or tutor. Edit, push, or delete. |
| Pushed Sets | Sets currently active for their class or learners. Completion rate per set. |
| Learner Progress | Per learner: which sets they have studied, their session scores, card mastery %. |
| Ratings | Average rating per card set from learners. Individual comments if provided. |

### 21.4.3 Parent View

| Section | Content |
|---|---|
| My Card Sets | Sets the parent created for their child. |
| Child’s Progress | Which card sets their child has studied, session scores, mastery progress. |
| Ratings | Whether their child rated a card set and what they said. |

## 21.5 Flashcard Rules & Behaviour Summary

| Rule | Detail |
|---|---|
| Learner cards are private | Cards a learner creates are only visible on their own profile. No one else can see them unless a sharing feature is added in a future phase. |
| Pushed cards are read-only for learners | Learners can study pushed cards but cannot edit or delete them. They can only rate them. |
| Cards are topic-linked | Every card set is linked to a topic in the platform’s content hierarchy. Cards created during a lesson are auto-linked to that topic. |
| Prompt limit | Maximum 2 flashcard creation prompts per study session. Prompts only appear at natural break points, never mid-question. |
| AI generation is draft-first | AI-generated cards are always shown to the creator for review before being pushed to any learner. No AI content reaches a learner without human approval. |
| Repeat curve resets per session | The within-session repeat queue resets each time a new session starts. Cross-session flagging persists until the card is mastered. |
| Mastery threshold | A card is marked Mastered after being answered correctly in 3 consecutive sessions. Mastered cards move to the back of rotation but are not removed. |
| Card set rating is optional | Learners are prompted to rate a pushed set after completing it. They can skip. Rating prompt only appears once per set. |

# 22. Roles, Hierarchy & Governance

Every learning space on both products has a Manager — the person in charge. The Manager controls who participates, what gets approved, and how the space runs. This is the governance layer that makes the platform flexible enough for private tutoring and structured enough for institutional school deployment.

**The central design principle: every space has at least one Manager. A space can have more than one Manager. Outside of EduSchool's fixed institutional hierarchy, the Manager role is chosen at sign-up — not assigned by the platform.**

## 22.1 The Sign-Up Question (EduTutor)

Before selecting a plan, every new EduTutor user is asked one question that determines their role, their dashboard, and their authority within the platform:

**How will you be using EduTutor? A - I am a learner managing my own learning | B - I am a parent managing my child's learning | C - I am a tutor managing my learners | D - I am a learner joining someone else's space**

A learning space in EduTutor is a private environment shared between a learner, optionally a tutor, and optionally a parent. Any combination of these can hold the Manager role. The space is created when the Manager registers and invites participants.

### 22.2.1 Configuration Options

| Configuration | Manager(s) | Participants | Use Case |
|---|---|---|---|
| Tutor-Managed | Tutor (sole Manager) or Tutor + Parent as co-Managers | Learner(s). Parent optionally linked for visibility. | Professional tutoring. Tutor runs the space. Parent observes or co-manages. Learner participates. Paid plans only. |
| Learner-Managed | Learner (sole Manager) or Learner + Parent as co-Managers | Tutor optionally added. Parent optionally linked. | Independent learner who finds their own tutor. Adult learner. Learner is in control. |
| Parent-Managed | Parent (sole Manager) or Parent + Learner as co-Managers | Child (learner). Tutor optionally added and approved by parent. | Younger learner whose parent drives the learning. Parent selects and approves the tutor. Learner participates. |
| Learner + Parent Only (No Tutor) | Parent-Manager or Learner-Manager or both as co-Managers | No tutor. Self-paced learning with parental oversight. | Free or paid tier. No tutor cost. Parent and/or learner manage the space together. |

### 22.2.2 Co-Manager Rules

- A space can have up to two co-Managers at any time.

- Co-Managers have equal authority within the space: both can invite, approve, remove participants, and manage sessions.
- A co-Manager can be added by the existing Manager from space settings at any time.
- Co-Managers cannot remove each other. Only the original Manager (the one who created the space) can remove a co-Manager.
- If the original Manager deletes their account, co-Manager status is promoted to sole Manager automatically.

## 22.3 EduTutor — Permission Matrix

Permissions in a learning space depend on whether a user holds the Manager role, a co-Manager role, or a Participant role in that space. ✓ = permitted, ✗ = not permitted, ■ = permitted if Manager or co-Manager.

| Action | Tutor (Manager) | Learner (Manager) | Parent (Manager) | Tutor (Participant) | Learner (Participant) | Parent (Observer) |
|---|---|---|---|---|---|---|
| Invite a tutor to the space | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| Invite a learner to the space | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| Approve a join request | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| Remove a participant | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| Add a co-Manager | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| Create and push assignments | ✓ | ✗ | ✗ | ✓ | ✗ | ✗ |
| Create and push flashcards | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Schedule a session | ✓ | ✓ | ✓ | ✓ | ✓ | ✗ |
| Approve a session booking | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| View all learner reports | ✓ | ✗ | ✓ | ✓ | ✗ | ✓ |
| View own progress only | ✗ | ✓ | ✗ | ✗ | ✓ | ✗ |
| Override AI mark | ✓ | ✗ | ✗ | ✓ | ✗ | ✗ |
| Upload lesson content | ✓ | ✗ | ✗ | ✓ | ✗ | ✗ |
| Change space approval settings | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |
| Delete the space | ✓ | ✓ | ✓ | ✗ | ✗ | ✗ |

## 22.4 EduTutor — Approval Flows

The Manager of a space configures approval settings when they set up their space. These settings determine who must approve before a session is confirmed or a new participant joins. Settings can be changed at any time from space settings.

### 22.4.1 Session Booking Approval

When anyone schedules a session, the approval chain fires before the session is confirmed:

| Approval Setting (set by Manager) | Who Receives the Approval Request | Session Confirmed When |
|---|---|---|
| Manager approval only | Manager receives the booking request. | Manager taps Approve. |
| All participants must approve | Every participant in the space receives the request. | All participants have tapped Approve. |
| Any one participant must approve | All participants notified. First to approve confirms it. | One participant taps Approve. |
| Auto-confirm | No approval needed. Session confirmed immediately on creation. | Instantly on creation by any Manager. |

Default setting: Manager approval only. Tutors working with younger learners are recommended to use All participants so both learner and parent confirm every session.

### 22.4.2 Join Request Flow

When a participant requests to join a space (e.g. a learner finds a tutor and requests to join their space), the following flow applies:

1. Participant submits a join request from the tutor's profile page or via an invite link.
2. All Managers of the space receive a notification: "[Name] has requested to join your space."
3. Any Manager can approve or decline. First Manager to act decides.
4. If approved: participant is added to the space and receives a welcome notification.
5. If declined: participant is notified politely. No reason is required.
6. If no action within 48 hours: request expires and participant is notified to try again.

### 22.4.3 Removing a Participant

Any Manager can remove any participant from the space at any time:

- Removed participant's access to the space ends immediately.
- Their personal progress data (lessons completed, answers, flashcards they created) is preserved on their own account.
- Content the removed participant created (e.g. a tutor's uploaded lessons) remains in the space unless the Manager explicitly deletes it.
- A learner under 18 who is a self-Manager can flag a request to remove their tutor. The platform prompts them to confirm, and if a parent is linked, the parent is also notified of the removal.

## 22.5 EduTutor — Free Tier Governance Rules

The Free tier has specific governance restrictions to reflect the product positioning: free access is for self-managed and parent-managed learning only. Tutor functionality requires a paid tutor package.

| Rule | Detail |
|---|---|
| No tutor on Free tier | A learner on the Free tier cannot be part of a tutor-managed space. They can upgrade at any time to join or create a tutor space. |
| Learner-managed Free space | A learner on Free can manage their own space, invite a parent for visibility, and study independently. No tutor participation. |
| Parent-managed Free space | A parent on Free can manage their child’s learning space with no tutor. Daily limits (3 lessons / 3 answers) apply to the learner. |
| Tutor requires paid package | A tutor must have an active Starter, Professional, Institution, or Enterprise package to manage a space, invite learners, or participate as a tutor in any space. |
| Learner joining a tutor’s space | When a learner joins a tutor’s space, the learner’s tier determines their daily limits. If the tutor’s package covers learner access (future enterprise feature), limits may be overridden by the tutor’s plan. |

## 22.6 EduSchool — Fixed Institutional Hierarchy

EduSchool uses a fixed hierarchy that mirrors how schools are actually structured. Unlike EduTutor where the Manager role is chosen at sign-up, EduSchool roles are assigned top-down. The School Admin is always at the top. Teachers manage their classes. Learners participate. Parents observe.

| Level | Role | Created By | Authority |
|---|---|---|---|
| 1 — Highest | School Admin | Platform Admin (on school sign-up) OR first teacher to register for the school (acting School Admin until a formal admin is assigned) | Full school authority. Adds and removes teachers. Views all classes, all learners, all reports. Approves platform-wide lesson sharing. Manages school subscription. |
| 1 — Equal | Co-School Admin | School Admin | Same authority as School Admin. Useful for deputy principal or HOD. School can have multiple co-admins. |
| 2 | Teacher | School Admin | Manages their own class(es). Adds and removes learners from their class. Links parents. Creates and pushes content. Views class reports. Cannot manage other teachers. |
| 3 | Learner | Teacher or School Admin | Participates in their class. Cannot self-enrol in EduSchool. No management authority. |
| 3 — Parallel | Parent | Teacher (sends invite) | Read-only portal. Sees their child’s progress only. No management authority in EduSchool. |

### 22.6.1 First Teacher Acting as School Admin

When a school signs up but the School Admin account has not yet been formally configured, the first teacher to register for that school is automatically granted temporary Acting School Admin status. This allows the school to get started immediately without waiting for a Platform Admin to manually create an admin account.

- Acting School Admin has full school admin permissions.
- They are clearly labelled “Acting Admin” in the platform until a formal School Admin is designated.
- The Platform Admin is notified that a new school has registered with an acting admin.
- The acting admin can promote themselves or another teacher to formal School Admin status.
- Once a formal School Admin is assigned, the acting admin label is removed and the user reverts to Teacher role (unless they were promoted to formal admin).

## 22.7 EduSchool — Permission Matrix

| Action | School Admin | Teacher | Learner | Parent |
|---|---|---|---|---|
| Add a teacher to the school | ✓ | ✗ | ✗ | ✗ |
| Remove a teacher from the school | ✓ | ✗ | ✗ | ✗ |
| Add a learner to a class | ✓ | ✓ | ✗ | ✗ |
| Remove a learner from a class | ✓ | ✓ | ✗ | ✗ |
| Link a parent to a learner | ✓ | ✓ | ✗ | ✗ |
| Push an assignment to a class | ✓ | ✓ | ✗ | ✗ |
| View all classes school-wide | ✓ | ✗ | ✗ | ✗ |
| View own class only | ✓ | ✓ | ✗ | ✗ |
| View own child’s progress only | ✗ | ✗ | ✗ | ✓ |
| Create and share lesson content | ✓ | ✓ | ✗ | ✗ |
| Approve platform-wide lesson sharing | ✓ | ✗ | ✗ | ✗ |
| Override AI mark | ✓ | ✓ | ✗ | ✗ |
| Create flashcards for class | ✓ | ✓ | ✗ | ✗ |
| Create personal flashcards | ✗ | ✗ | ✓ | ✓ |
| Manage school subscription | ✓ | ✗ | ✗ | ✗ |
| Generate school-wide reports | ✓ | ✗ | ✗ | ✗ |
| Generate class reports | ✓ | ✓ | ✗ | ✗ |

## 22.8 Role Comparison Across Both Products

The same person may use both EduTutor and EduSchool with different roles in each. A teacher who uses EduSchool for their class might also use EduTutor privately as a tutor for individual students outside school hours. Their roles and permissions are entirely independent across the two products.

| Concept | EduTutor | EduSchool |
|---|---|---|
| Who is the Manager? | Chosen at sign-up. Can be learner, parent, or tutor. Flexible. | Fixed by hierarchy. School Admin is always the top authority. |
| Can there be co-Managers? | Yes. Up to two co-Managers per space. Equal authority. | Yes. Multiple co-School Admins. Teachers co-manage classes if assigned. |
| Can a learner self-enrol? | Yes. Learner can register independently and manage their own space. | No. Learner is enrolled by teacher or school admin only. |
| Parent role | Observer by default. Can be promoted to co-Manager by the space Manager. | Observer only. Cannot be promoted. No management authority. |
| Approval flows | Configurable per space by the Manager. Can require all parties to approve. | Fixed. Teachers add learners directly. No approval required. |
| Free tier governance | No tutor on Free. Learner and parent can manage a space on Free. | Not applicable. Schools subscribe institutionally. |
| First user authority | First user sets their role at sign-up. | First teacher to register becomes acting School Admin automatically. |

## 22.9 EduTutor Sign-Up Flow Diagram

The diagram below shows the complete EduTutor sign-up flow from the role selection question through to dashboard activation, including the under-18 parent linking recommendation.

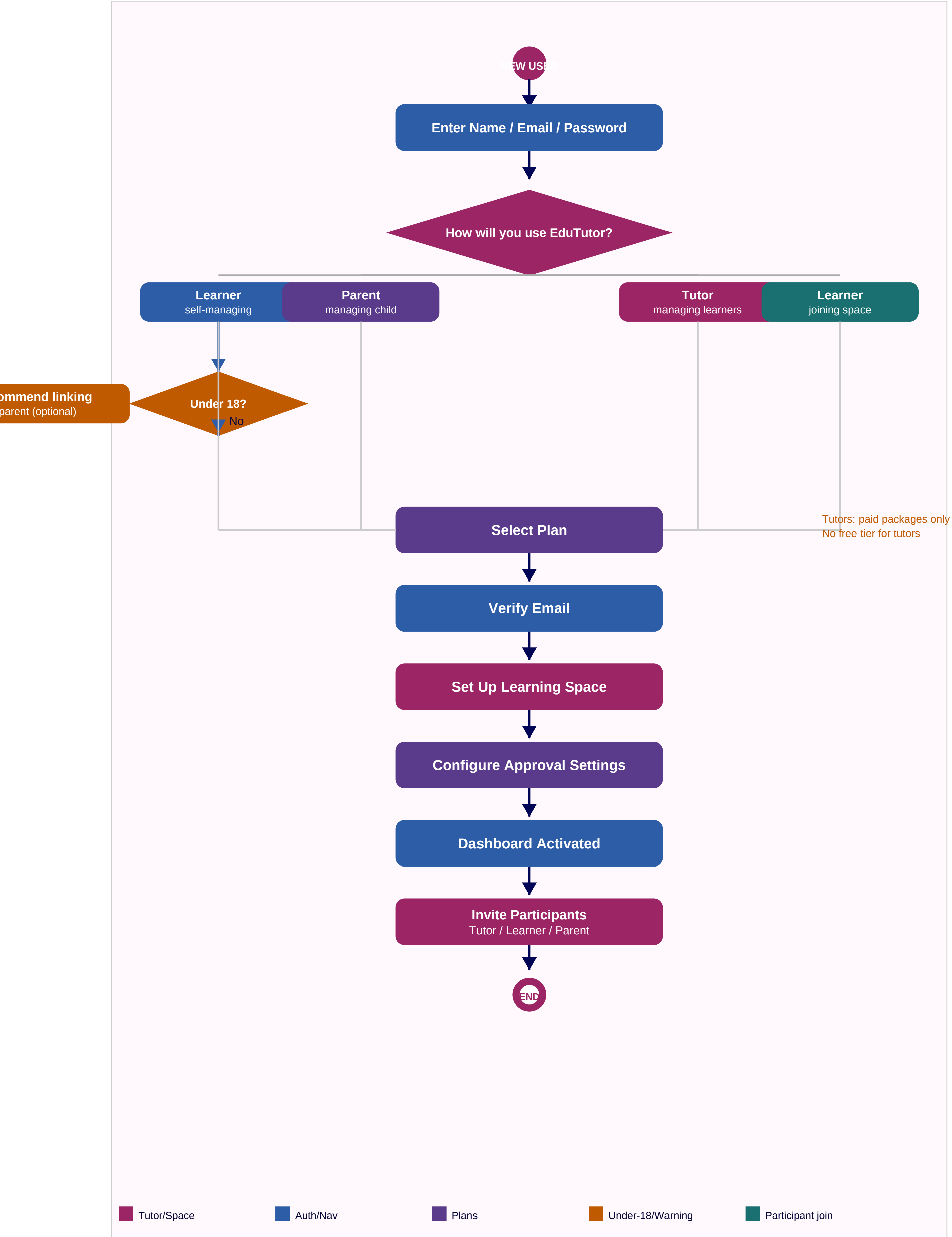

EW USE
Enter Name / Email / Password
How will you use EduTutor?
Learner
self-managing
Parent
managing child
Tutor
managing learners
Learner
joining space
Under 18?
No
ommend linking
parent (optional)
Select Plan
Tutors: paid packages only
No free tier for tutors
Verify Email
Set Up Learning Space
Configure Approval Settings
Dashboard Activated
Invite Participants
Tutor / Learner / Parent
END
Tutor/Space
Auth/Nav
Plans
Under-18/Warning
Participant join

*Figure 5: EduTutor sign-up and role assignment flow*

# 23. Goals, Reminders & Persistent Alerts

Goals and reminders form the proactive layer of both platforms. A teacher or tutor sets a goal for their learner or class. The platform then sends reminders to drive the learner toward that goal. If the learner does not respond, the reminder escalates — it does not disappear after one notification. The alert continues until the goal is met or the educator acknowledges it.

**An unattended alert is not a failed notification. It is a signal that a learner needs intervention. The platform keeps alerting until a human acts.**

## 23.1 Who Can Set Goals

| Role | Can Set Goals For | Goal Scope |
|---|---|---|
| Teacher (EduSchool) | Their entire class, or an individual learner within their class. | Class goal applies to all learners. Individual goal overrides the class goal for that learner. |
| Tutor (EduTutor) | One or more of their assigned learners. | Per-learner goal. Each learner in the tutor's space can have their own goal. |
| Parent (EduTutor) | Their child only. | Per-child goal. Parent sets in Learner-Parent Mode or alongside a tutor in Tutor-Learner Mode. |
| School Admin (EduSchool) | The entire school, a specific grade, or a specific class. | School-wide or grade-wide goals cascade down to class level. Teacher can override for their class. |

## 23.2 Goal Types

Goals are measurable targets tied to platform activity. Every goal type has a corresponding metric the platform already tracks, so goal progress is automatic — no manual logging required.

| Goal Type | Example | How Progress Is Tracked |
|---|---|---|
| Daily lessons | Complete at least 1 video lesson per day. | Platform counts lessons watched per calendar day. Resets at midnight SAST. |
| Daily questions | Answer at least 5 questions per day. | Platform counts questions submitted per calendar day. |
| Weekly topic completion | Complete at least 1 full topic per week (all lessons + consolidation passed). | Platform checks consolidation pass status per topic per 7-day window. |
| Daily study time | Spend at least 20 minutes actively using the platform per day. | Platform measures active session time (not idle). Resets at midnight SAST. |

| | | |
|---|---|---|
| Assignment completion | Complete the pushed assignment before a specified deadline. | Platform checks assignment status against deadline timestamp. |
| Session attendance | Attend all scheduled sessions in the week. | Platform checks attendance confirmation and session join records. |
| Flashcard study | Study flashcards at least 3 times per week. | Platform counts flashcard study sessions per 7-day window. |

Goals can be set with a start date and an optional end date (e.g. for a school term). A goal with no end date runs until manually stopped by the educator.

## 23.3 Setting a Goal — Teacher & Tutor Workflow

### 23.3.1 Teacher Sets a Class Goal (EduSchool)

1. Teacher navigates to their class dashboard and taps “Set Class Goal.”
2. Teacher selects goal type (e.g. Daily questions), target value (e.g. 5 questions), and reminder time (e.g. send reminder at 6pm if goal not met by then).
3. Teacher sets escalation behaviour: how many unanswered reminders before the teacher is alerted and a flag appears on the learner’s tile.
4. Teacher optionally sets a goal end date (e.g. end of term).
5. Goal is saved and immediately active for all learners in the class. Each learner sees their goal progress bar on their dashboard.
6. Teacher can override the class goal for a specific learner (e.g. set a lower target for a struggling learner, higher for an advanced one).

### 23.3.2 Tutor Sets a Learner Goal (EduTutor)

1. Tutor opens the learner’s profile and taps “Set Goal.”
2. Tutor selects goal type, target, and reminder time.
3. Tutor sets escalation behaviour.
4. Goal is saved. Learner sees their goal on their dashboard.
5. If a parent is linked, the parent also sees the goal in their portal.

A learner can have multiple active goals simultaneously (e.g. a daily question goal and a weekly topic goal). Each goal has its own reminder schedule and escalation path.

## 23.4 Reminder Schedule

When a goal’s deadline approaches and the learner has not met their target, the platform sends reminders. Reminders are not one-and-done — they follow a schedule and escalate if not acted upon.

### 23.4.1 Standard Reminder Flow

| Reminder Stage | Timing | Recipient | Message Example |
|---|---|---|---|
| First reminder | At the time set by the educator (e.g. 6pm if daily goal not yet met). | Learner | “You haven’t completed your daily goal yet: 5 questions. You’ve done 2. Tap to continue.” |
| Second reminder | 2 hours after the first reminder if goal still not met. | Learner | “Still 3 questions to go to hit your goal today. You’ve got time — tap to start.” |
| Third reminder | 1 hour before midnight if goal still not met. | Learner | “Last chance to hit your goal today. Complete 3 more questions before midnight.” |
| Goal missed | At midnight if goal was not met. | Learner + Educator | Learner: “You missed your goal today. No problem — start fresh tomorrow.” Educator: streak broken, flag raised on dashboard. |

### 23.4.2 Escalation — Persistent Alert Until Action Is Taken

If a learner misses their goal on consecutive days without the educator acknowledging it, the alert escalates. The alert does not disappear until a human acts. This is the core of the accountability loop — the platform does not silently give up.

| Consecutive Misses | What Happens |
|---|---|
| 1 miss | Goal streak broken. Learner notified. Educator sees a yellow flag on the learner’s tile or dashboard entry. |
| 2 consecutive misses | Educator receives a direct alert: “[Learner] has missed their daily goal 2 days in a row.” Flag on dashboard turns amber. |
| 3 consecutive misses | Educator receives a priority alert. If a parent is linked, parent also receives: “[Child] has not been meeting their learning goal for 3 days. Consider checking in.” Flag turns red on dashboard. |
| 5+ consecutive misses | Platform sends a summary escalation to the educator: “[Learner] has missed their goal 5 days in a row. This learner may need direct intervention.” Alert remains on the educator’s dashboard until they take an action (send a nudge, book a session, adjust the goal, or manually dismiss). |
| Alert dismissed without action | Educator can dismiss an alert. Dismissed alerts are logged. If the learner continues missing goals after dismissal, the escalation restarts from day 1. |

### 23.4.3 Educator Acknowledgement Actions

An escalated alert is cleared when the educator takes one of the following actions from the alert directly:

- Send a nudge to the learner — platform records the nudge against the alert.
- Adjust the goal — if the target was too high, educator reduces it. Alert clears.
- Book a session — schedules a session to address the issue directly. Alert clears.
- Notify parent — sends a parent notification and logs it against the alert. Alert clears.
- Manually dismiss — educator acknowledges but takes no platform action. Logged with a timestamp.

Every action taken on an escalated alert is logged in the learner's activity record. School Admin can see all unacknowledged alerts across all classes at any time.

## 23.5 Goal Progress — What Each Person Sees

### 23.5.1 Learner View

| Element | What It Shows |
|---|---|
| Goal progress bar | On the learner's dashboard: "3/5 questions done today." Bar fills as the learner works. Green when goal is met. Resets daily/weekly. |
| Streak counter | Number of consecutive days/weeks the learner has met their goal. Displayed prominently on the dashboard as a motivator. |
| Goal history | Calendar view showing which days the goal was met (green) and which were missed (red). Learner can see their own pattern over time. |
| Reminder toggle | Learner can adjust the time they receive reminders (e.g. move from 6pm to 7pm) but cannot disable goal reminders set by their educator. They can only disable reminders for goals they set themselves. |

### 23.5.2 Teacher View (EduSchool)

| Element | What It Shows |
|---|---|
| Class goal overview | Summary bar on class dashboard: X/35 learners met their goal today. Percentage complete across the class. |
| Individual flags | Learner tiles on the live session view and class dashboard show yellow/amber/red flags based on consecutive missed goals. |
| Unacknowledged alerts | A dedicated "Alerts" panel on the teacher's dashboard listing all learners with escalated alerts. Sorted by urgency (most consecutive misses first). Alert remains until teacher acts. |
| Goal history per learner | Teacher can tap any learner to see their full goal history: which days they met the goal, which they missed, and what actions the teacher took. |
| School Admin view | School Admin sees all unacknowledged alerts across all classes. Can reassign an alert to a specific teacher if no action has been taken. |

### 23.5.3 Tutor View (EduTutor)

| Element | What It Shows |
|---|---|
| Per-learner goal card | On the tutor's monitoring dashboard: each learner shows their current goal, today's progress, and current streak. At a glance without opening individual profiles. |
| Alert panel | Unacknowledged escalated alerts listed at the top of the tutor's dashboard. Alert persists until tutor acts. |

| | |
|---|---|
| Session prep integration | When a tutor opens their session prep view before a session, any active escalated goals are shown at the top: "[Learner] has missed their daily question goal 3 days in a row." This feeds directly into session planning. |

### 23.5.4 Parent View

| Element | What It Shows |
|---|---|
| Daily goal summary | Parent portal shows: "[Child] met their goal today: 5/5 questions completed." Or: "[Child] missed their goal today: 2/5 questions completed." |
| Streak visible | Parent can see the child's current streak and goal history calendar. |
| Escalation notification | Parent receives a notification at 3 consecutive misses (as described in 23.4.2). Notification includes a direct link to the child's progress in the portal. |
| Goal set by educator | Parent can see the goal that was set by the teacher or tutor but cannot change it. Parent can add their own supplementary goal (e.g. a flashcard study goal) that runs alongside the educator's goal. |

## 23.6 Goal Rules & Behaviour Summary

| Rule | Detail |
|---|---|
| Educator goals cannot be disabled by learner | A learner can snooze a single reminder by 1 hour but cannot turn off goal reminders set by their teacher, tutor, or parent. |
| Multiple goals can be active simultaneously | A learner can have a daily question goal and a weekly topic goal active at the same time. Each has its own reminder and escalation path. |
| Class goal vs individual goal | If a teacher sets a class goal and then an individual goal for a specific learner, the individual goal overrides the class goal for that learner only. |
| Goal cascade in EduSchool | School Admin sets a school-wide or grade-wide goal. It cascades to all classes. Teachers can override for their class. Teachers can override for individual learners. |
| Escalation resets on action | If the educator takes any acknowledgement action, the consecutive miss counter resets. It starts again from day 1 if misses resume. |
| Goal streaks are motivational only | Streaks are displayed to motivate learners but do not affect platform access, daily limits, or progress through content. |
| Goals are logged permanently | All goals, reminder events, missed days, and educator actions are stored in the learner's activity log and visible in reports. |
| Quiet hours respected | Reminders do not fire outside the user's configured quiet hours (default: no reminders between 9pm and 7am). |

## 8.9 Pricing Viability Review

A per-learner infrastructure cost analysis against the new pricing model reveals one tier that requires attention before going to market.

| Scenario | Infrastructure Cost/Learner | Revenue/Learner | Gross Margin | Status |
|---|---|---|---|---|
| Tutor Solo package (5 learners, 20 answers/day) | ~R5.34 | R199 / 5 = R39.80 | ~87% | ✓ Healthy |
| Tutor Small package (15 learners, 40 answers/day) | ~R8.46 | R349 / 15 = R23.27 | ~64% | ✓ Healthy |
| Tutor Medium package (30 learners, 80 answers/day) | ~R15.46 | R599 / 30 = R19.97 | ~23% | ■ Tight — review |
| EduSchool (any size, 60 ans/day, R28/learner) | ~R12.00 | R28.00/learner | ~57% | ✓ Healthy — see Section S3 |
| Tutor Medium (30 learners, repriced) | ~R17.16 | R39.97/learner | ~57% | ✓ Healthy — repriced |
| Tutor Large (50 learners, repriced) | ~R17.00 | R39.98/learner | ~58% | ✓ Healthy — repriced |
| Tutor Studio (100 learners, repriced) | ~R16.93 | R39.99/learner | ~58% | ✓ Healthy — repriced |
| Learner self-managed Free tier | ~R2.21 | R0 | Loss leader (intentional) | ✓ Acceptable — conversion funnel |
| Learner self-managed Premium (R149/month) | ~R15.46 | R149 | ~90% | ✓ Healthy |

### 8.9.1 Pricing Floor Policy

Effective from this revision, every package across both products is priced to guarantee a minimum 55% gross margin. No package is offered below this floor. This required two changes:

- EduSchool: moved from banded pricing (R12–R18/learner) to a flat R28/learner/month rate with a 60 AI-answer/day limit. Full detail in Section S3.
- EduTutor Medium, Large, and Studio tutor packages repriced to R1,199 / R1,999 / R3,999 respectively (~R40/learner), up from R599 / R899 / R1,299. Full detail in Section T2.2.
- Solo, Small, and all self-managed learner tiers were already above 55% and remain unchanged.
- All pricing figures are TBD — these are indicative. Run a market sensitivity test before fixing final prices. The 55% floor is the non-negotiable constraint; the absolute price points can move based on market feedback as long as the floor is maintained.

# 24. Device Strategy

Both EduTutor and EduSchool are built for web and mobile simultaneously, not sequentially. This is a deliberate architectural decision. Building in parallel ensures the backend API is designed from day one to serve both surfaces with the same data contracts, and that every feature decision accounts for which channel delivers the best experience — rather than defaulting everything to web and retrofitting mobile later.

> **An email reminder is right for a web user not checking the app. A push notification is right for a mobile user who has the app installed. If you build web first, you default everything to email. You only discover the problem when mobile arrives and it's painful to unpick. Build in parallel. Design for both channels from the start.**

## 24.1 Device Coverage

| Platform | How Delivered | Who Uses It | Notes |
|---|---|---|---|
| Web (all devices) | Responsive web app via browser (Chrome, Safari, Edge, Firefox). | All users on PC, Mac, tablet, or any smartphone browser. | Primary delivery surface. Full feature access. Works on any device with a modern browser. No installation required. |
| Android app | Native Flutter app distributed via Google Play Store. | Android smartphone and tablet users. | Full native experience. Push notifications. Camera access for answer photo uploads and Phase 2 video. Offline access to flashcards and downloaded content. |
| iOS app | Native Flutter app distributed via Apple App Store. | iPhone and iPad users. | Same Flutter codebase as Android. Separate App Store build. Apple Developer license required (~R1,850/year). Push notifications via APNs. |
| Windows desktop | Progressive Web App (PWA) installable via Chrome or Edge. Flutter desktop app optional in Phase 2. | PC users in schools or home offices. | PWA installs to taskbar and runs like a desktop app. No separate build required at launch. Flutter desktop can be added in Phase 2 if needed. |
| Mac desktop | Web via Safari or Chrome. Flutter desktop app optional in Phase 2. | Mac users (teachers, tutors, parents on MacBook). | Web experience is excellent on Mac. No separate build needed at launch. |

### 24.2 Why Not Android-Only for Native?

Android has ~75% smartphone market share in South Africa, which would justify Android-only for MVP. However:

- Teachers and tutors in urban private schools skew heavily toward iPhone and MacBook. This is EduSchool's primary early market.

- Flutter produces both Android and iOS from the same codebase. The incremental cost of shipping iOS alongside Android is low — mostly the App Store submission process and Apple developer license.
- A school that standardises on iPads (common in SA private schools) needs an iOS app for the class deployment to work.

Decision: build Android and iOS in parallel from the same Flutter codebase. The marginal effort is small. The market coverage is complete.

## 24.3 What Lives Where — Feature-to-Channel Map

Not every feature works equally well on every surface. This table defines the right channel for each key feature and why. Designing this at the start prevents retrofitting later.

| Feature | Best on Web | Best on Mobile App | Reason |
|---|---|---|---|
| Lesson video playback | ✓ | ✓ | Both work well. Mobile benefits from offline download capability (Phase 2). |
| Answering practice questions | ✓ | ✓ | Both work. Typed answers easier on web keyboard. Photo upload of handwritten work better on mobile camera. |
| Push notifications (session reminders, goal alerts) | ✗ (email fallback) | ✓ | Native push notifications are more reliable and immediate on mobile. Web push is unreliable on iOS Safari. Email is the right fallback for web-only users. |
| Email notifications (daily summaries, weekly reports) | ✓ | ✗ (push preferred) | Email suits web users who check their inbox. Mobile users prefer push for time-sensitive alerts and email for digests. |
| Session scheduling & attendance confirmation | ✓ | ✓ | Both. The 15-minute session reminder is a push notification — critical on mobile. Web email fallback covers users not on mobile. |
| Teacher live session view | ✓ (large screen preferred) | ✓ (monitoring on the go) | Large screen better for seeing all 35 learner tiles at once. Mobile useful if teacher walks the room and monitors their phone. |
| Flashcard study | ✓ | ✓ (preferred) | Flashcards are naturally mobile — short study sessions during spare moments. Mobile experience is primary. |
| Answer photo upload (handwritten work) | ✗ (file upload fallback) | ✓ | Mobile camera is the natural input. Web users can upload a photo file but it's less seamless. |
| In-session video & whiteboard (Phase 2) | ✓ | ✓ | Both. WebRTC works in browser and in Flutter app. Mobile benefits from native camera quality. |

| | | | |
|---|---|---|---|
| Session recording playback | ✓ (large screen preferred) | ✓ | Both. Review on large screen for study. Mobile for quick revision. |
| Admin / School Admin dashboard | ✓ (primary) | ✗ (limited view only) | Complex dashboards with tables and charts suit large screens. Mobile gets a simplified summary view only. |
| Goal progress bar & streak | ✓ | ✓ (primary) | Learner-facing. Mobile is where learners interact daily. Progress bar on home screen is a key motivator. |
| Offline content access | ✗ | ✓ (Phase 2) | Web requires connectivity. Mobile Flutter app can cache lessons and flashcards for offline use when data is limited. |

## 24.4 Notification Channel Decision Matrix

Every notification in the system is designed for both channels simultaneously. This table defines the primary and fallback channel for each notification type, determined at design time.

| Notification | Mobile App (Push) | Web User (Email) | Both (Push + Email) |
|---|---|---|---|
| Session reminder — 15 min before | ✓ Primary | Fallback only | If both channels active |
| Session starting now | ✓ Primary | Fallback only | If both channels active |
| Assignment pushed to class | ✓ Primary | ✓ Also sent | Always both |
| AI marking feedback ready | ✓ Primary | Digest only | Push instant, email in daily digest |
| Goal not met — first reminder | ✓ Primary | Fallback only | If both |
| Goal not met — escalated (3+ days) | ✓ Primary | ✓ Also sent | Always both — escalation warrants email |
| Learner inactive 48 hours (to parent) | ✓ Primary | ✓ Also sent | Always both |
| Daily activity summary (to parent) | Daily digest push | Daily digest email | User chooses preferred channel |
| Session summary after session ends | ✓ Primary | ✓ Also sent | Always both |
| New flashcard set pushed | ✓ Primary | Digest only | Push instant |
| Consolidation passed | ✓ Primary | Digest only | Push instant |
| Weekly goal report (to parent/tutor) | Weekly push | ✓ Primary | Email primary for weekly digest |

| | | | |
|---|---|---|---|
| Lesson request fulfilled | ✓ Primary | ✓ Also sent | Always both |
| Package/subscription renewal reminder | ✓ Primary | ✓ Primary | Always both — financial |

Users without the mobile app installed receive all push-primary notifications via email automatically. No configuration required. Once they install the app and enable notifications, push takes over and email reverts to digest-only for non-critical alerts.

# 25. Technology Stack — Portfolio-Aligned Selection

This stack is chosen with a second goal alongside building the product: every tool selected either fills a gap between Herve's CV and his demonstrable portfolio, or directly matches tooling named in target job descriptions (Expleo, CyberSentriq, QA Supervisor roles). ShopFlow already demonstrates GitHub Actions, Semgrep, SonarCloud, and Docker. This project is deliberately structured to demonstrate the CV-claimed tools that ShopFlow does not yet show: Azure DevOps (Pipelines + Boards), Jenkins, BrowserStack, Selenium, TestNG, Page Object Model, REST Assured, and Postman — alongside filling the explicitly stated Playwright gap.

**Two parallel QA tracks, by design: a modern TypeScript track (Playwright + Vitest) for the product itself, and a classic enterprise SDET track (Java + Selenium + TestNG + Page Object Model + REST Assured) testing the same running application. One project, two demonstrable skill sets — exactly matching the dual stacks seen across SA QA job descriptions.**

### 25.0 CV-to-Portfolio Gap Analysis

| CV Tool | In ShopFlow? | In EduPlatform? | How |
|---|---|---|---|
| Selenium / Java | ✗ No | ✓ Yes | qa-java module — Selenium WebDriver smoke suite |
| TestNG | ✗ No | ✓ Yes | qa-java module — TestNG runner + reporting |
| Page Object Model | ✗ No | ✓ Yes | qa-java module — POM classes per page |
| REST Assured | ✗ No | ✓ Yes | qa-java module — API contract tests vs NestJS API |
| Postman | ✗ No | ✓ Yes | Postman collections + Newman in CI |
| BrowserStack | ✗ No | ✓ Yes | Selenium suite run on BrowserStack cloud grid |
| Docker | ✓ Yes | ✓ Yes | Continued — all services containerised |
| GitHub Actions | ✓ Yes | ✓ Yes | Continued — fast PR checks |
| Jenkins | ✗ No | ✓ Yes | Self-hosted, runs qa-java suite nightly |
| Azure DevOps | ✗ No | ✓ Yes | Azure Boards (backlog) + Azure Pipelines (release) |
| Semgrep / SonarCloud | ✓ Yes | ✓ Yes | Continued — extended to Java module |
| Playwright (gap) | ✗ No | ✓ Yes | Primary E2E framework for the web app |

## 25.1 Development Stack

The application stack itself remains the modern TypeScript/Flutter approach from the original design — this is what makes the product buildable and is a separate concern from the QA/DevOps portfolio goal above.

| Layer | Technology |
|---|---|
| Backend | NestJS (Node.js + TypeScript), Prisma ORM, PostgreSQL via Supabase |
| Web Frontend | Next.js 14, Tailwind CSS, Zustand, TanStack Query |
| Mobile | Flutter (Android + iOS + desktop) |
| Monorepo | Nx + Turborepo |
| Queue / Cache | Redis + BullMQ |

## 25.2 QA Stack — Two Tracks

### 25.2.1 Track A — Modern TypeScript Track (the product's own tests)

This track tests the product the way a modern engineering team would, and fills the stated Playwright gap directly.

| Layer | Tool | Purpose | CV Relevance |
|---|---|---|---|
| Unit / Integration | Vitest | Fast unit and integration tests for NestJS and React components. | New — modern equivalent of JUnit, increasingly requested alongside Jest in JDs. |
| E2E (web) | Playwright (TypeScript) | End-to-end browser automation. Auto-waiting, parallel execution, trace viewer. | Fills the stated Playwright gap. Appears in nearly every modern SDET JD alongside Selenium. |
| API mocking | Mock Service Worker (MSW) | Intercepts API calls in frontend tests without a live backend. | New — supporting tool, speeds up Track A. |
| Performance | k6 | Load testing the AI marking queue and API under concurrent learners. | New — modern alternative to JMeter; mention both in interviews. |

### 25.2.2 Track B — Classic Enterprise SDET Track (qa-java module)

A dedicated module — qa-java/ — lives inside the Nx monorepo as a standalone Maven Java project. It tests the same running EduPlatform application (web UI and API) using the stack listed on Herve's CV. This directly answers the common interview question: 'Can you show me an example of this?'

| Tool | Purpose | Where It's Used in EduPlatform |
|---|---|---|
| Selenium WebDriver (Java) | Browser automation for UI test cases. | Smoke suite: registration, login, lesson navigation, question submission, AI feedback display. |
| Page Object Model | Maintainable test architecture — one class per page/component. | LoginPage, DashboardPage, TopicPage, QuestionPage, FlashcardPage classes mirror the Next.js app. |

| | | |
|---|---|---|
| TestNG | Test runner, assertions, grouping, parallel execution, HTML reporting. | Suites grouped by feature (auth, content, AI marking, flashcards). XML suite files define run order. |
| REST Assured | API contract and integration testing in Java. | Validates the NestJS API: auth, content CRUD, AI marking response shape, subscription endpoints. |
| BrowserStack | Cloud grid for cross-browser and cross-device execution. | Selenium suite runs against BrowserStack (Chrome, Safari, Edge, real Android/iOS devices). |
| Postman + Newman | API collection design and CI-runnable smoke tests. | A maintained Postman collection covering every endpoint, run via Newman as a pre-deploy check. |

Track B runs on a different cadence to Track A: Playwright runs on every PR (fast feedback). The Selenium/TestNG/REST Assured suite via BrowserStack runs nightly against staging — mirroring how SA enterprises actually structure CI vs nightly regression.

### 25.2.3 Test Strategy Summary

| Test Type | Track | Tooling | Trigger |
|---|---|---|---|
| Unit / Component | A | Vitest | Every commit |
| E2E (fast) | A | Playwright | Every PR |
| API smoke | B | Postman/Newman | Every PR (fast subset) |
| Cross-browser/device E2E | B | Selenium + TestNG + BrowserStack | Nightly against staging |
| API contract/regression | B | REST Assured + TestNG | Nightly against staging |
| Load test | A | k6 | Weekly / pre-release |

## 25.3 DevOps Stack

Source code stays on GitHub — this preserves the GitHub Codespaces mobile development workflow established earlier. CI/CD and project management deliberately route through Azure DevOps and Jenkins, since these are CV-claimed tools without portfolio evidence.

| Tool | Role in This Project | CV Relevance |
|---|---|---|
| GitHub (repo host) | Source control. Codespaces for phone-based development. PRs and code review. | Already on CV and ShopFlow — kept for the mobile workflow it enables. |
| Azure Boards | Sprint backlog, work items, kanban board. Linked to GitHub repo via Azure Boards GitHub integration. | On CV, no portfolio evidence — now demonstrated with a real maintained backlog. |
| Azure Pipelines | Release pipeline: builds Nx affected projects, runs qa-java suite, deploys to staging/production. Source is the GitHub repo. | On CV, no portfolio evidence — now demonstrated with a working YAML pipeline. |

| | | |
|---|---|---|
| Jenkins (self-hosted) | Nightly job: triggers the qa-java BrowserStack suite against staging, publishes TestNG/Allure HTML report. | On CV, no portfolio evidence — now demonstrated with a real nightly job + reports. |
| GitHub Actions | Fast PR checks only: lint, typecheck, Vitest, Playwright (Track A), Postman/Newman smoke. | Already on CV and ShopFlow — kept for fast feedback. |
| SonarCloud + Semgrep | Code quality and SAST, applied to TypeScript app and qa-java module. | Already on CV and ShopFlow — extended to a second language (Java). |
| Docker | Containerises NestJS API, Redis, Postgres (local dev), and the Jenkins agent. | Already on CV and ShopFlow — extended usage. |
| Doppler | Secrets management across GitHub Actions, Azure Pipelines, and Jenkins. | New — relevant given three CI systems need shared secrets. |
| Sentry | Error monitoring for NestJS API and Next.js web app in staging/production. | New — commonly listed in SDET/DevOps JDs alongside CI tooling. |

### 25.3.1 Why Three CI Systems Is a Feature, Not Clutter

Running GitHub Actions, Azure Pipelines, and Jenkins on the same project might look redundant on a real production team — but for a portfolio project built to demonstrate breadth, it is the point. Each system has a clear, non-overlapping job:

- GitHub Actions — fast feedback on every PR (lint, unit tests, Track A E2E).
- Azure Pipelines — the release pipeline (build, package, deploy), demonstrating Azure DevOps end-to-end including Boards integration.
- Jenkins — scheduled nightly regression (Track B), demonstrating a classic on-prem-style QA pipeline.

In an interview, this becomes: ‘I built one project but ran it through three CI ecosystems to show I can work in whichever your team uses — GitHub Actions for speed, Azure DevOps for release management and backlog, Jenkins for scheduled QA regression.’

### 25.3.2 Repository Structure Addition

| Path | Contents |
|---|---|
| qa-java/ | Standalone Maven project. src/test/java contains: page objects (POM), TestNG suite XML files, Selenium WebDriver tests (BrowserStack-configured), and REST Assured API test classes. Has its own pom.xml — intentionally separate from the Nx/npm toolchain to mirror a real polyglot repo. |
| postman/ | Postman collection (JSON) covering every API endpoint, plus environment files for local/staging/production. Run via Newman in GitHub Actions. |
| .azure-pipelines/ | azure-pipelines.yml defining the release pipeline (build → qa-java → deploy). |
| jenkins/ | Jenkinsfile defining the nightly BrowserStack regression job. |

### 25.3.3 Recommended Build Order

| Order | Item | Rationale |
|---|---|---|
| 1 | Nx workspace + NestJS + Next.js (Day 1 PDF) | Foundation — nothing else works without it. |
| 2 | GitHub Actions: lint/typecheck/Vitest/Playwright on PR | Fast feedback established early, fills Playwright gap immediately. |
| 3 | Postman collection + Newman in GitHub Actions | Quick win — demonstrates Postman with minimal setup. |
| 4 | Azure Boards backlog (move epics from GitHub Issues) | Demonstrates Azure DevOps project management once epics exist. |
| 5 | qa-java: Selenium + POM + TestNG (local browsers first) | Build and prove locally before adding cloud grid complexity. |
| 6 | qa-java: add REST Assured API tests | Extends the same module — low marginal effort. |
| 7 | BrowserStack: point qa-java at it | Swap local WebDriver for remote driver — small config, big portfolio value. |
| 8 | Jenkins: nightly job running qa-java against staging | Requires staging to exist — do after core features deployed. |
| 9 | Azure Pipelines: release pipeline to staging/production | Final piece — ties Boards + Pipelines together end-to-end. |
| 10 | Doppler across all three CI systems | Do once all three CI systems exist and need shared secrets. |

# 26. Payment Gateway Strategy

Both EduTutor and EduSchool are subscription-based platforms. Payment processing must support recurring monthly billing, multiple SA payment methods, card tokenisation, and failed payment retries. All gateway options below are South African, ZAR-native, and developer-friendly.

## 26.1 Payment Gateway Options

| Gateway | Transaction Fee | Monthly Fee | Recurring Billing | Best For | Developer Exp |
|---|---|---|---|---|---|
| PayFast | ~3.5% + R2.00 (cards). ~2% (Instant EFT). | R0 (no monthly fee) | Subscription tokens. Requires billing engine (BullMQ handles this in NestJS). | MVP and early production. Fastest setup. Widest SA payment method support: cards, Instant EFT, SnapScan, Zapper, Mobicred. | Excellent. Well-document API. Sandbox a NestJS SDK av |
| Peach Payments | ~3.2% (cards). Custom rates at R200k+/month. | Custom (volume-based) | Native recurring billing and tokenisation built in. Best subscription support in SA. | Scale (R200k+/month). Schools paying large contracts. Native recurring billing without a separate billing engine. | Strong. REST A Webhook supp 90-second real payouts (2026) |
| Stitch | Custom (API-first pricing) | Custom | Instant EFT and bank-linked payments. Strong API. No card-on-file recurring. | Instant EFT-first checkout. High-growth startups. Not ideal as primary subscription gateway. | Excellent DX. A Best for EFT flo alongside a car gateway. |
| Ozow | ~1.5% (Instant EFT only) | R0 | No native card recurring. EFT-only. | Supplementary EFT option alongside PayFast or Peach. Lower fee for EFT payments. | Good. REST A well with PayFa |

## 26.2 Recommended Gateway Strategy

| Phase | Primary Gateway | Supplementary | Rationale |
|---|---|---|---|
| MVP (Launch) | PayFast | None | Zero monthly fee. Fastest setup (hours not days). Widest payment method support. Subscription tokens work with BullMQ billing engine in NestJS. Covers cards, Instant EFT, SnapScan, Zapper, and Mobicred in one integration. |

| | | | |
|---|---|---|---|
| Growth (R50k +/month) | PayFast | Ozow (EFT) | Add Ozow for Instant EFT at ~1.5% vs PayFast's 2% EFT rate. Saves ~0.5% on every EFT transaction. At R50k/month EFT volume that's R250/month saving. |
| Scale (R200k +/month) | Peach Payments | Ozow (EFT) | Switch primary card processing to Peach for lower card rates, native recurring billing (removes need for BullMQ billing engine), and real-time 90-second payouts. Negotiate custom rate at this volume. |

### 26.3 How Subscription Billing Works in the Platform

Subscriptions are managed inside the NestJS backend, not delegated entirely to the gateway. This gives full control over billing logic: upgrades, downgrades, prorations, failed payment retries, and seat-count-triggered plan changes.

1. Learner or tutor selects a plan and enters payment details at checkout.
2. PayFast tokenises the card and returns a subscription token to the NestJS backend.
3. NestJS stores the token against the user account (never the raw card details).
4. A BullMQ recurring job fires on the billing date each month.
5. NestJS calls PayFast API with the stored token to charge the subscription amount.
6. On success: subscription renewed, limits reset, user notified.
7. On failure: retry logic fires (day 1, day 3, day 7 after failure). User notified at each retry. After 3 failures: account downgraded to Free tier with a grace period notification.
8. On upgrade/downgrade: proration calculated, charge or credit applied on next billing date.
9. On school contract (EduSchool): manual invoice generated monthly. School pays via EFT. School Admin sees payment status in billing dashboard.

School contracts (EduSchool) are invoiced manually via EFT rather than card-on-file. This is standard practice for institutional SA buyers who pay via purchase order. PayFast or Peach are used for individual EduTutor subscriptions only.

### 26.4 Payment Gateway Cost Per Transaction

Every subscription payment incurs a gateway fee. This must be factored into the per-learner cost model. The table below shows the effective gateway cost per learner per month at each price point using PayFast card rates (~3.5% + R2.00):

| Subscription | Monthly Price | PayFast Fee (3.5% + R2) | Net Revenue | Gateway Fee % |
|---|---|---|---|---|
| Learner Free | R0 | R0 | R0 | N/A |
| Learner Basic | R59 | R4.07 | R54.93 | 6.9% |
| Learner Standard | R99 | R5.47 | R93.53 | 5.5% |
| Learner Premium | R149 | R7.22 | R141.78 | 4.8% |
| Tutor Solo (R199/5 learners) | R199 | R8.97 | R190.03 | 4.5% |

| | | | | |
|---|---|---|---|---|
| Tutor Small (R349/15) | R349 | R14.22 | R334.78 | 4.1% |
| Tutor Medium (R599/30) | R599 | R22.97 | R576.03 | 3.8% |
| Tutor Studio (R1,299/100) | R1,299 | R47.47 | R1,251.53 | 3.7% |

At scale, switching to Peach Payments (~3.2% rate) saves approximately 0.3% per transaction. At R200,000/month in card revenue that is R600/month saved. Not material at launch but worth switching at scale.

# 27. Complete Cost Per Learner Model

This section combines all three cost layers for the first time: infrastructure (AI, video, storage, hosting), tooling (tech stack licensing and services), and payment gateway fees. The result is the true cost to serve one learner per month and the real gross margin at each pricing tier.

## 27.1 Tooling Cost (Monthly Fixed + Variable)

These are the monthly costs of the technology stack itself, independent of learner count. Most tools are free at small scale and only incur cost at growth milestones.

| Tool | Free Tier / Cost | Paid Trigger | Paid Cost |
| --- | --- | --- | --- |
| Supabase (DB + Auth + Realtime) | Free up to 50,000 MAUs | 50,000 MAUs exceeded | Pro: $25/month (R460) |
| Railway (backend hosting) | $5/month (hobby) | High traffic | Pro: $20/month (R368) |
| Coolify (self-hosted PaaS) | $0 software. VPS ~$10/month (R184) | Traffic growth | $20-40/month VPS upgrade (R368-736) |
| Redis (BullMQ + caching) | Included in Railway hobby | High queue volume | Upstash Redis: $0.20/100k commands |
| Sentry (error monitoring) | Free: 5,000 errors/month | Production errors exceed free | Team: $26/month (R478) |
| Grafana Cloud (metrics) | Free: 10,000 metrics | Large infrastructure | Pro: $8/month (R147) |
| Checkly (synthetic monitoring) | Free: 10 checks | More production checks needed | Starter: $30/month (R552) |
| Doppler (secrets management) | Free: unlimited secrets, 3 projects | Team features needed | Team: $7/month/user (R129/user) |
| Nx Cloud (build caching) | Free: 500 CI runs/month | Large team / high CI volume | Pro: $25/month (R460) |
| Storybook (component docs) | Free (self-hosted) | N/A | R0 |
| Allure Report | Free (self-hosted) | N/A | R0 |
| Renovate Bot | Free on GitHub | N/A | R0 |
| DeepSeek API (AI marking) | 5M free tokens on signup | Immediately after free grant | Pay per use (see Section 8) |
| Bunny.net (video CDN) | No free tier. Pay per use from R0.18/GB | Immediately | Pay per use |
| Cloudinary (images) | Free: 25GB storage + 25GB bandwidth | 25GB exceeded | Plus: $89/month (R1,638) |

| | | | |
|---|---|---|---|
| Apple Developer License (iOS) | N/A | Required for App Store | $99/year = R153/month |
| Google Play Developer (Android) | N/A | One-time registration | $25 once (R460) |
| Domain names x2 | N/A | Annual | ~R37/month each |

### 27.1.1 Monthly Fixed Platform Cost at Launch

| Cost Item | Monthly (ZAR) |
|---|---|
| Railway hobby (backend + Redis) | R92 |
| Cloudinary free tier | R0 |
| Supabase free tier | R0 |
| Sentry free tier | R0 |
| Doppler free tier | R0 |
| VPS for Coolify (optional at launch) | R184 |
| Apple Developer License | R153 |
| Google Play (amortised) | R4 |
| Domain names x2 | R74 |
| Grafana free tier | R0 |
| Checkly free tier | R0 |
| TOTAL FIXED AT LAUNCH | R507/month |

R507/month is your minimum platform cost before any learners are on the platform. This covers both EduTutor and EduSchool running simultaneously.

## 27.2 True Cost Per Learner Per Month (All-In)

The table below shows the complete per-learner cost including: infrastructure (AI + video + storage + hosting share), tooling (fixed cost amortised across learner base), and payment gateway fee. Fixed tooling is amortised at 100 active learners for these calculations.

| Tier | Infra Cost | Tooling Share | Gateway Fee | TOTAL COST | Revenue | Net Margin |
|---|---|---|---|---|---|---|
| Free (self-managed) | R2.21 | R5.07 | R0 | R7.28 | R0 | Loss leader |
| Learner Basic (R59) | R5.34 | R5.07 | R4.07 | R14.48 | R59 | ~75% |
| Learner Standard (R99) | R8.46 | R5.07 | R5.47 | R19.00 | R99 | ~81% |
| Learner Premium (R149) | R15.46 | R5.07 | R7.22 | R27.75 | R149 | ~81% |

| | | | | | | |
|---|---|---|---|---|---|---|
| Tutor Solo per learner (R199/5) | R5.34 | R1.01 | R1.79 | R8.14 | R39.80 | ~80% |
| Tutor Small per learner (R349/15) | R8.46 | R0.34 | R0.95 | R9.75 | R23.27 | ~58% |
| Tutor Medium per learner (R1,199/30) | R15.46 | R0.17 | R1.53 | R17.16 | R39.97 | ~57% |
| Tutor Large per learner (R1,999/50) | R15.46 | R0.10 | R1.44 | R17.00 | R39.98 | ~58% |
| Tutor Studio per learner (R3,999/100) | R15.46 | R0.05 | R1.42 | R16.93 | R39.99 | ~58% |
| EduSchool (60 ans/day, R28/learner) | R12.00 | R0.03 | R0 | R12.03 | R28.00 | ~57% |

**Every package and band now guarantees at least 55% gross margin. Tooling share drops further as learner count grows (R507 fixed cost becomes R0.05/learner at 10,000 learners), pushing actual margins slightly above the figures shown here over time. No package is below the 55% floor.**

### 27.3 Cost at Different Scale Points

The table below shows total monthly platform cost and blended margin at three growth milestones, assuming a mixed learner base (40% Free, 30% Basic, 20% Standard, 10% Premium for EduTutor self-managed; plus tutor-covered learners):

| Scale | Active Learners | Monthly Infra + Tooling | Monthly Revenue (est.) | Blended M |
|---|---|---|---|---|
| Pre-revenue (testing) | 0 | R507 fixed | R0 | N/A — fixe |
| Early traction | 100 | ~R2,600 | ~R6,500 | ~60% |
| Growth | 500 | ~R10,500 | ~R32,000 | ~67% |
| Scale | 2,000 | ~R36,000 | ~R130,000 | ~72% |
| With 1 EduSchool (300 learners) | +R3,609 infra | +R8,400 revenue | ~R8,400 vs R3,609 | ~57% — he |

### 27.4 Pricing Floor Compliance Summary

Following the repricing in Sections S3 and T2.2, every package and band across both products meets or exceeds the 55% gross margin floor. The table below summarises the final position:

| Product | Package/Band | Final Price | Margin | Status |
|---|---|---|---|---|
| EduTutor Learner | Free | R0 | Loss leader (intentional) | ✓ By design |
| EduTutor Learner | Basic | R59/month | ~75% | ✓ |
| EduTutor Learner | Standard | R99/month | ~81% | ✓ |

| | | | | |
|---|---|---|---|---|
| EduTutor Learner | Premium | R149/month | ~81% | ✓ |
| EduTutor Tutor | Solo (5) | R199/month | ~80% | ✓ |
| EduTutor Tutor | Small (15) | R349/month | ~58% | ✓ |
| EduTutor Tutor | Medium (30) | R1,199/month | ~57% | ✓ Repriced |
| EduTutor Tutor | Large (50) | R1,999/month | ~58% | ✓ Repriced |
| EduTutor Tutor | Studio (100) | R3,999/month | ~58% | ✓ Repriced |
| EduSchool | Any enrolment | R28/learner | ~57% | ✓ Repriced |

All pricing remains TBD for final market launch — these are indicative figures. The 55% margin floor is the structural constraint that must hold; absolute price points can be adjusted based on market sensitivity testing as long as the floor is maintained for every package and band.

# 7. Future Considerations

### Both Products

- Flashcard sharing: learners can optionally share their own card sets with classmates or the platform community.
- Flashcard import/export: export card sets as PDF or CSV for offline use.
- Multi-language support: isiZulu, Afrikaans, Sesotho, and other official SA languages.
- Offline mode for learners with limited or no data access.
- Social login (Google, Microsoft) for faster registration.
- Gamification: streaks, badges, leaderboards for learner motivation.
- AI lesson plan suggestions generated from class or learner performance data.

### EduTutor

- Parent portal for EduTutor (currently planned as future feature).
- Tutor marketplace: learners can discover and subscribe to tutors on the platform.
- Tutor earnings dashboard if marketplace model is introduced.

### EduSchool

- In-app parent-teacher communication (currently read-only portal).
- Integration with SA school management systems (EMIS, D6, etc.).
- Department / HOD reporting layer above teacher, below school admin.
- Lesson quality rating: teachers rate shared lessons from colleagues.

End of Document — EduTutor & EduSchool System Design v10.0